

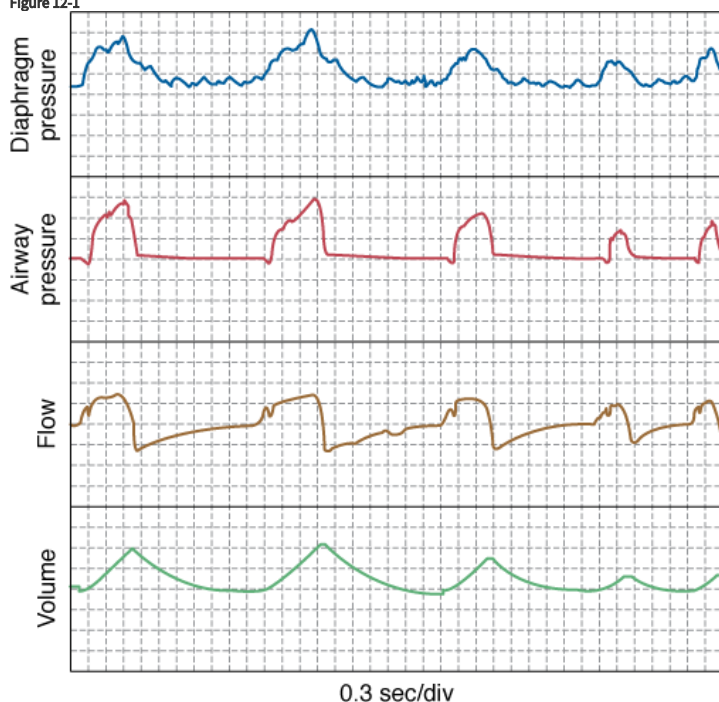
Chapter 12. Proportional-Assist Ventilation

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Proportional-Assist Ventilation: Introduction

Proportional-assist ventilation (PAV) is a form of synchronized ventilator support in which the ventilator generates pressure in proportion to *instantaneous* patient effort (Fig. 12-1).¹ The ventilator simply amplifies inspiratory efforts. Unlike other modes of partial support, there is no target flow, tidal volume, or ventilation or airway pressure. Rather, PAV's objective is to allow the patient to comfortably attain whatever ventilation and breathing pattern his or her control system desires.¹ The main operational advantages of PAV are automatic synchrony with inspiratory efforts and adaptability of the assist to changes in ventilatory demand (Fig. 12-1).

Figure 12-1



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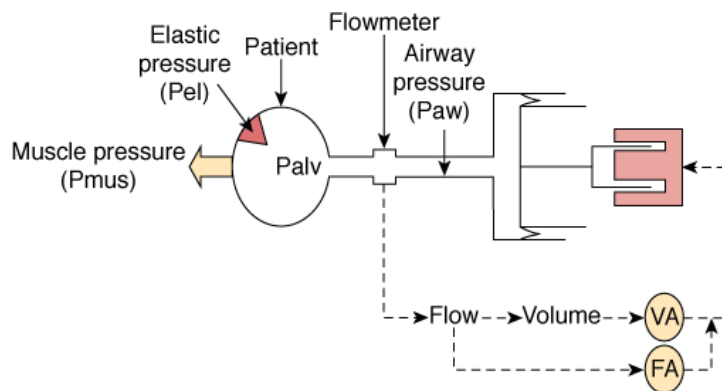
Relationship between assist provided (airway pressure) and independently measured diaphragmatic pressure in proportional-assist ventilation. Note that amplitude and duration of assist correspond to amplitude and duration of inspiratory efforts.

Basic Principles and Algorithms: How Can a Ventilator Deliver Pressure in Proportion to Patient Effort Without Directly Measuring Effort?

A simple PAV delivery system illustrates how this happens (Fig. 12-2).² Alveoli and chest wall are represented as an elastic compartment that opposes expansion. Elastic recoil pressure (P_{el} ; hatched arrow in Fig. 12-2) is a function of how much lung volume deviates from passive functional residual capacity (FRC) and the stiffness of the system: $P_{el} = V \times E$, where V is volume above FRC and E is respiratory system elastance. In a passive system, P_{el} increases alveolar pressure as the lung is artificially inflated. During assisted ventilation, inspiratory muscles are active. These muscles decrease alveolar pressure by an amount corresponding to their pressure output (P_{mus}) (Fig. 12-2). At any instant, alveolar pressure (P_{alv}) is the difference between P_{el} ($V \times E$), which tends to increase it, and P_{mus} , which tends to decrease it:

$$P_{alv} = (V \times E) - P_{mus} \quad (1)$$

Figure 12-2



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Diagram illustrating how a PAV delivery system generates pressure in proportion to effort. The gas-delivery system consists of a freely moving piston pressurized by a fast-acting motor. Force exerted by motor is a function of flow and volume leaving the ventilator. A stronger effort results in greater reduction in alveolar pressure (P_{alv}), drawing more gas from the piston and resulting in more assist. If the volume-assist (VA) and flow-assist (FA) components are set to the same fraction of elastance and resistance, respectively, the pressure generated becomes proportional to effort (P_{mus}). See text.

The elastic compartment is connected to the external tubing via airways and the endotracheal tube. The ventilator controls pressure at the external airway (P_{aw}). Air flows into the lungs when P_{aw} exceeds P_{alv} . Flow is a function of the difference between P_{aw} and P_{alv} (resistive pressure) and the resistance of the intervening tubing (R). Thus

$$\text{Flow} = (P_{aw} - P_{alv})/R \quad (2)$$

Substituting equation (Eq.) 1 for P_{alv} in Eq. 2 and rearranging, we get

$$\text{Flow} \times R = P_{aw} - (V \times E) + P_{mus}$$

or

$$P_{mus} + P_{aw} = \text{flow} \times R + V \times E \quad (3)$$

This equation simply states that the distending force is the sum of patient-generated (P_{mus}) and ventilator-generated (P_{aw}) pressures and that this distending force is opposed by the sum of resistive pressure drop (resistive pressure, or $\text{flow} \times R$) and elastic recoil pressure (P_{el} , or $V \times E$).

The gas-delivery system in Figure 12-2 is a freely moving piston pressurized by a fast-responding linear motor. This arrangement emphasizes that PAV gas-delivery systems must allow rapid and free flow of gas in response to changes in alveolar pressure. Flow and volume leaving the ventilator are measured. The gains of the flow and volume signals are adjustable by separate amplifiers: flow assist (FA) and volume assist (VA). The summed output of the two amplifiers is the input to the motor. Thus, the ventilator's pressure output is a function of instantaneous flow and volume that left the ventilator since triggering.

With this arrangement (see Fig. 12-2), a greater effort (more reduction in alveolar pressure) will draw more gas from the ventilator. This, in turn, results in more assist. This provides a positive relation between effort and assist but does not per se cause the assist to be proportional to instantaneous effort. Proportionality is achieved through customized adjustment of the FA and VA gains. The basis for these adjustments is as follows:

FA is the assist pressure per unit flow (in cm H₂O/L/s). These are resistance (R) units. If FA is 50% of R, the ventilator provides 50% of the resistive pressure (i.e., 50% of $\text{flow} \times R$, Eq. 3). At 80% R, the ventilator assumes 80% of resistive work, and so on. Likewise, setting VA gain to 50% of E causes the ventilator to assume 50% of elastic pressure (i.e., 50% of $V \times E$, Eq. 3), and so on. The total assist (P_{aw}) is the sum of the flow and volume assists:

$$P_{aw} = \% \text{flow} \times R + \% V \times E \quad (4)$$

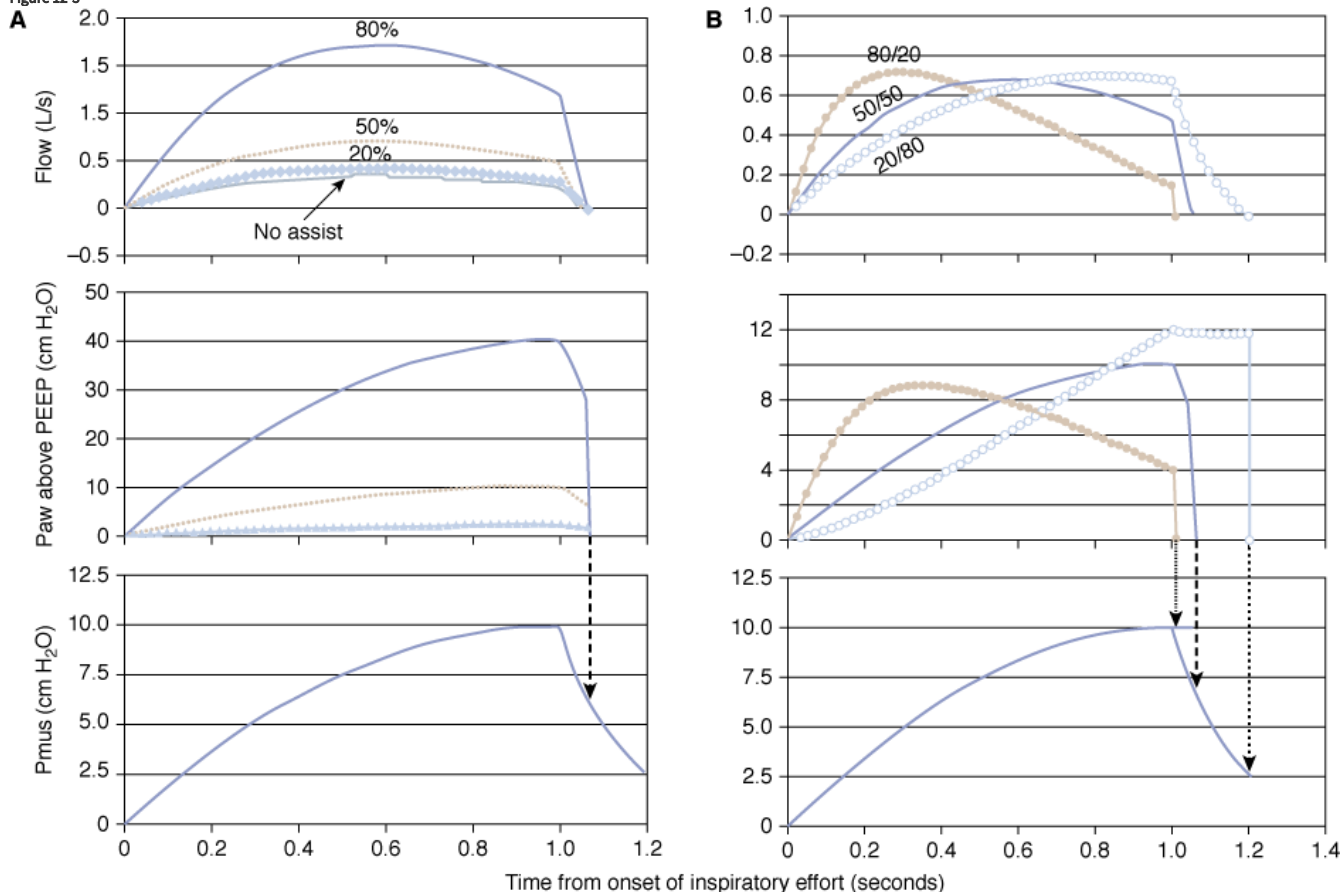
During the inspiratory phase, volume rises progressively, peaking at end inspiration. By contrast, flow peaks in early to middle inspiration and falls later. Thus, the relative contributions of resistive and elastic pressures vary considerably during the inspiratory phase. If the same percent is used for both components, total assist (P_{aw}) represents the same percent of total pressure regardless of the relative contribution of each. Percent assist then is constant throughout. If different percent values are used for FA and VA, however, total assist (P_{aw}) will represent a different percent of total applied pressure at different times. When percent assist (ventilator's contribution) is constant throughout inspiration, patient's percent contribution (i.e., 100 – percent assist) is also constant throughout and the relationship between P_{aw} and P_{mus} (i.e., proportionality becomes constant) as given by

$$\text{Proportionality} = \text{percent assist} / (100 - \text{percent assist})$$

Thus, at 50% assist, proportionality is 1.0; P_{aw} equals P_{mus} throughout. At 80% assist, proportionality is 4, and so on. Under these conditions, the shapes of the P_{mus} and P_{aw} waveforms are identical, and the decline in P_{mus} at end inspiration is associated with a decline in P_{aw} , ensuring synchrony (Fig. 12-

3A). By contrast, if percent assist varies through inspiration (such as by using different percent values for FA and VA), proportionality between P_{aw} and P_{mus} is no longer constant, and the shapes of the two waveforms differ (Fig. 12-3B).

Figure 12-3



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Model simulation showing the impact of using equal (A) versus unequal (B) percent assist for the flow and volume components. **A.** Three assist levels are shown: 20%, 50%, and 80%. When the same percent assist is applied to both components, the shape of the assist (airway pressure [P_{aw}]) is identical to that of muscle pressure (P_{mus}), but the proportionality is different ($P_{aw}/P_{mus} = 0.25, 1.0, \text{ and } 4.0$, respectively). Note also that flow reaches zero (end of mechanical inspiration) at the same time during the declining phase of P_{mus} at all assist levels, including the no-assist situation (*top panel*). **B.** Flow assist is 80% of resistance, and volume assist is 20% of elastance (*solid dots*). Note that the assist (P_{aw}) is more aggressive early in inspiration and terminates sooner relative to the balanced assist (*50/50 line*). A relatively greater volume assist (*open circles*) offers less assist early while cycling off is delayed.

Physiologic Effects

Relevant Physiologic Principles

These principles are discussed first because they not only help to explain PAV's reported effects, but also make it possible to gain useful insights from a patient's responses to this mode; reference 3 provides more details. Responses to PAV may be mediated by changes in blood-gas tensions (chemical factors) or through modification of nonchemical sources of respiratory drive. Chemical responses are highly predictable, whereas the others are not. What happens, therefore, depends on what sources of respiratory drive are operative at the time of application.

Sources of Respiratory Drive

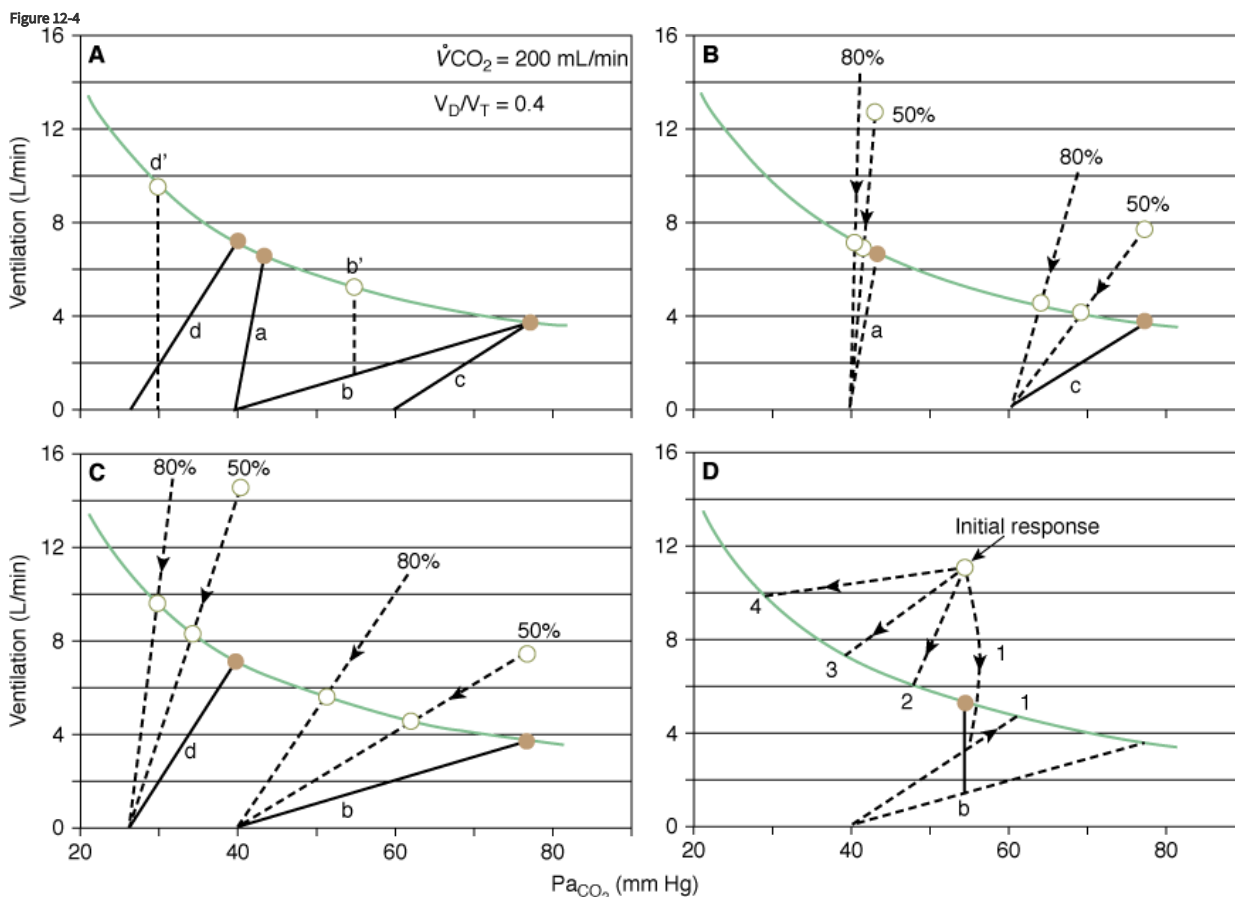
During sleep and anesthesia, chemical factors are the sole source of respiratory drive; artificially reducing the partial pressure of carbon dioxide (P_{CO_2}) under these conditions abolishes respiratory efforts.⁴⁻⁶ Furthermore, in these states, respiratory muscle responses to changes in load are mediated exclusively via changes in blood-gas tensions.⁷ Conversely, in alert individuals, other sources of respiratory drive exist; it is very difficult to produce apnea by assisted ventilation despite marked hypocapnia.⁸⁻¹¹ These drive inputs, collectively called the *consciousness factor*,⁴ presumably arise from behavioral centers and from respiratory mechanisms that operate only during consciousness (e.g., nonchemical load-compensatory mechanisms⁷). Patients who require mechanical ventilation cover a wide spectrum of levels of consciousness. Therefore, it is difficult to make general conclusions about

their drive inputs. The next few sections describe what should happen in response to PAV if respiratory output were driven solely by chemical factors. Deviation from this expected behavior then might be attributed to nonchemical factors.

Determinants of $\dot{V}_E$ and P_{aCO_2} in the Absence of Nonchemical Drive Sources

Without Assist

A steady state in P_{CO_2} and minute ventilation ($\dot{V}_E$) can occur only if pulmonary carbon dioxide (CO_2) removal (the product of alveolar ventilation [$\dot{V}_E$] and alveolar CO_2 concentration [F_{ACO_2}]) equals the CO_2 produced by the tissues ($\dot{V}_{CO_2}$). At a given $\dot{V}_{CO_2}$, if ventilation is increased artificially, F_{ACO_2} , and hence P_{aCO_2} , must decrease before a steady state is reached. Accordingly, at a given $\dot{V}_{CO_2}$, there is an inverse relationship between $\dot{V}_E$ and P_{aCO_2} in the steady state—the metabolic hyperbola.¹² The actual equation is $P_{aCO_2} = 0.86 [\dot{V}_{CO_2} / \dot{V}_E (1 - V_D/V_T)]$,¹² where V_D/V_T is the dead-space-to-tidal-volume ratio. Figure 12-4A illustrates this relationship.



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Effect of proportional assist on steady-state ventilation ($\dot{V}_E$) and P_{aCO_2} . A metabolic hyperbola is shown for a subject with CO_2 production ($\dot{V}_{CO_2}$) of 200 mL/min and a dead-space-to-tidal-volume ratio (V_D/V_T) of 0.4. **A.** Unassisted breathing. The ventilatory responses to CO_2 in four subjects are shown: (a) normal, (b) acute hypercapnic failure, (c) chronic hypercapnic failure, and (d) acute nonhypercapnic failure with metabolic acidosis. In each case, the x intercept of the ventilatory response is the apneic threshold. The apneic threshold is shifted up in subject c and down in subject d. The intersection of ventilatory response line and the hyperbola gives steady-state values for $\dot{V}_E$ and P_{aCO_2} . Vertical dotted lines represent additional nonchemical inputs that cause ventilation to be higher and P_{aO_2} to be lower than in their absence. **B and C.** Effect of 50% and 80% assist in the four subjects. In each case the ventilatory response slope doubles with 50% assist and increases fivefold at 80%. Open circles off the hyperbola are immediate responses not consistent with steady state. Open circles on the hyperbola are the steady-state values with assist. Note that the magnitude of change in $\dot{V}_E$ and P_{aCO_2} is a function of the difference between unassisted P_{aCO_2} (solid circle) and the apneic threshold. **D.** Response to proportional assist in the presence of nonchemical input (vertical solid line). Note that the response is highly unpredictable (see text).

Ventilation increases progressively as a function of P_{CO_2} .¹³ The slope of the response depends on the sensitivity of chemoreceptors (the sensory arm) and the effectiveness of the motor arm in generating ventilation. Thus, for a given chemoreceptor response, ventilatory response is less if respiratory muscles are weaker or mechanics are abnormal. Without nonchemical drive sources, there is a P_{aCO_2} below which apnea develops,⁴⁻⁶ the apneic threshold (AT; see Fig. 12-4).

Four subjects with different disorders are illustrated in Figure 12-4A. For each subject, there is only one possible steady state, namely, the point of intersection of the CO_2 response line and the hyperbola.¹³ At this point, pulmonary CO_2 elimination equals tissue CO_2 production.

Line *a* represents a normal subject. Ventilatory response is normal (2.2 L/min/mm Hg), and AT is 40 mm Hg. The intersection point is at a P_{aCO_2} of 43 mm Hg and a $\dot{V}_E$ of 6.6 L/min. The difference between unassisted (no assist [NA]) steady-state P_{CO_2} and AT (i.e., $\Delta P_{CO_2, NA-AT}$) is very small (3 mm Hg). Line *b* represents a patient with severe acute hypercapnic respiratory failure. AT is the same, but the ventilatory response to CO_2 is greatly depressed because of abnormal mechanics. The intersection point is 76 mm Hg at a $\dot{V}_E$ of 3.8 L/min. $\Delta P_{CO_2, NA-AT}$ is very large (36 mm Hg). Line *c* describes a patient with chronic respiratory failure. Ventilatory response to CO_2 is depressed because of abnormal mechanics and/or weak respiratory muscles. In this case, however, AT is higher. $\Delta P_{CO_2, NA-AT}$ is larger than normal (16 mm Hg), but for the same steady-state P_{aCO_2} , it is lower than in the patient represented by line *b*. Line *d* represents a patient with a reduced ventilatory response to CO_2 (0.6 L/min/mm Hg) because of abnormal mechanics or weak muscles but in whom the apneic threshold is low, for example, because of concomitant metabolic acidosis.¹⁴ Steady-state P_{CO_2} and $\dot{V}_E$ are near normal, but $\Delta P_{CO_2, NA-AT}$ is large.

Expected Response to Proportional-Assist Ventilation Application

With PAV, respiratory motor output is amplified by an amount that is related to percent assist (see Fig. 12-3A). Within the linear range of the pressure-volume and pressure-flow relationships, the amplification of pressure results in a corresponding amplification of ventilation (see Fig. 12-3A). Thus, the net effect of PAV is to increase the slope of the ventilatory response to CO_2 (as well as to the partial pressure of oxygen [P_{O_2}] and pH). For simplicity, we will assume that P_{CO_2} is the only stimulus and that resistance and elastance are constant in the tidal volume range. Under these conditions, at 50% assist, delivered assist equals P_{mus} , and the combined pressure (patient + ventilator) is twice P_{mus} . Accordingly, the slope of the CO_2 response is doubled. At 80% assist, P_{aw} is four times P_{mus} (see Fig. 12-3A), and pressure output is amplified by a factor of five. The ventilatory response should increase nearly fivefold, and so on.

In the normal subject (line *a*), 50% assist doubles the CO_2 response (see Fig. 12-4B). $\dot{V}_E$ immediately doubles (*upper open circle*). Pulmonary CO_2 output now exceeds $\dot{V}_{CO_2}$; P_{aCO_2} falls. Respiratory efforts decrease, and $\dot{V}_E$ follows along the new CO_2 response line until the metabolic hyperbola. A steady state is now possible. The same would happen at higher percent assist. Because steady-state values must be above the AT, and $\Delta P_{CO_2, NA-AT}$ is very small, $\dot{V}_E$ and P_{aCO_2} cannot change much. Furthermore, because $\dot{V}_E$ hardly changes, virtually all the assist (P_{aw}) is used to reduce respiratory motor output, and percent reduction in P_{mus} is similar to percent assist.

A similar analysis for the patient with chronic respiratory failure (line *c* in Fig. 12-4B) shows that at 50% assist, P_{aCO_2} decreases by 7 mm Hg and $\dot{V}_E$ increases by 10%, whereas at 80% assist, P_{aCO_2} decreases by 13 mm Hg and $\dot{V}_E$ increases by 20%. Because $\dot{V}_E$ increased significantly, the decrease in respiratory muscle output (P_{mus}) is less than percent assist. For example, at 50% assist, P_{mus} is contributing 50% to a higher $\dot{V}_E$ (110%). The decrease in P_{mus} is 45% instead of 50%. Because the difference between P_{aCO_2} at 80% assist and AT is now very small, increasing assist beyond 80% would have little further effect even though P_{aCO_2} is still high. Thus, when AT is high, it is not possible to acutely normalize P_{aCO_2} using PAV (or any other strictly assist mode).

In the patient with severe acute hypercapnic failure (line *b* in Fig. 12-4C), the changes are even greater. At 50% and 80% assist, P_{aCO_2} decreases by 14 and 25 mm Hg, respectively, and $\dot{V}_E$ increases by 23% and 50%. Even at 80% assist, the difference between P_{aCO_2} and AT is still large, and further reduction in P_{aCO_2} is possible. Because $\dot{V}_E$ increases substantially, the decrease in P_{mus} is much less than percent assist (39% and 70%, for the 50% and 80% assist, respectively).

Finally, in the patient represented by line *d* in Figure 12-4C, increasing PAV assist results in progressive hypocapnia. By normalizing mechanics, the effect of the concomitant metabolic acidosis is now exposed.

In summary, *in the absence of nonchemical drive sources*, whether and by how much ventilation and P_{aCO_2} change following institution of a given percent assist are determined by the unassisted ventilatory response to CO_2 , the apneic threshold, and the position of the metabolic hyperbola. These determine

the difference between unassisted P_{aCO_2} and AT. Because the more $\dot{V}_E$ increases, the less P_{mus} decreases, the same three factors determine the extent to which the assist is used to increase ventilation versus decrease muscle output.

Effect of Nonchemical Drives on Response to Proportional-Assist Ventilation

The action of nonchemical drive inputs can be viewed as additive with chemical drive. They cause ventilation to be higher at a given P_{aCO_2} than if chemical drive were the sole source (points b' and d' in Fig. 12-4A). Total drive is made up of a CO_2 -sensitive component and a component that reflects consciousness-related reflexes and unpredictable behavioral influences.

Figure 12-4D illustrates how these inputs may modify the response to PAV. Without nonchemical influences, would be 76 mm Hg. Because of the nonchemical input (*solid vertical line*), however, steady-state $\dot{V}_E$ is 5.5 L/min at a P_{aCO_2} of 55 mm Hg. A 50% assist results in an immediate increase in $\dot{V}_E$ to 11 L/min (both components are amplified). P_{aCO_2} must fall. What happens then depends on the response of the nonchemical component. At one extreme, it may disappear (e.g., the patient may fall asleep when assisted). $\dot{V}_E$ would fall to the new CO_2 response line (*diagonal dashed line*). Should $\dot{V}_E$ at this point be below the hyperbola, P_{aCO_2} and $\dot{V}_E$ will rise along the CO_2 response line, reaching the hyperbola at point 1. Here the assist is followed by an increase in P_{aCO_2} , but the patient is working much less. If the nonchemical stimulus remains the same, P_{aCO_2} and $\dot{V}_E$ will decrease along a path parallel to the CO_2 response line, meeting the hyperbola at point 3. Here, there is no longer a CO_2 stimulus, and ventilation is sustained by the twice-amplified nonchemical influence. An intermediate value (point 2) may result if the nonchemical influence partially decreases. Finally, at the other extreme, the patient may become agitated with the assist, increasing the nonchemical stimulus, and this, when amplified, may reduce P_{aCO_2} to very low levels (point 4). It is clear that with nonchemical stimuli (alert individuals), the ventilatory response to PAV is theoretically unpredictable.

Reported Physiologic Responses

Respiratory Muscle Output

PAV resulted in significant reduction in muscle output in all studies where this was tested.^{15–30} Typically, P_{mus} decreased by 30% to 45% at 50% assist and by 55% to 70% at 80% assist. The less-than-expected reduction is caused by (a) differences between assumed and actual E and R (see “[Accuracy and Stability of Respiratory Mechanics Values](#)” below), (b) imperfect delivery by the ventilator (see “[Ventilator Response Time](#)” below), (c) dynamic hyperinflation (see “[Dynamic Hyperinflation](#)” below) or nonlinearity in the pressure–volume relationship (see “[Nonlinearity in the Pressure–Volume Relationship within the Tidal Volume Range](#)” below), and (d) an associated increase in $\dot{V}_E$ (see “[Ventilation and Partial Pressure of Carbon Dioxide](#)” below). As indicated earlier, when some of the assist is used to increase $\dot{V}_E$, less is used to reduce muscle output.

Ventilation and Partial Pressure of Carbon Dioxide

Application of PAV to normal sleeping subjects results in an immediate increase in tidal volume (V_T) and $\dot{V}_E$, which decrease over several breaths to near-baseline levels.¹⁷ Steady-state responses are minimal,^{6,17} and the decrease in end-tidal CO_2 tension ($P_{ET}CO_2$) is very modest (approximately 3 mm Hg^{6,17}). These results are consistent with a small difference between unassisted P_{aCO_2} and AT secondary to a normal ventilatory response to CO_2 (subject represented by line a in Fig. 12-4B).

By contrast, in experienced, awake resting normal subjects, PAV results in an important increase in and a more pronounced reduction in P_{aCO_2} , and the decrease in respiratory muscle output is only modest.¹⁸ P_{aCO_2} generally decreases below the apneic threshold (e.g., 30 ± 5 mm Hg¹⁸), reflecting the presence of nonchemical drive inputs that fail to be eliminated. PAV applied to inexperienced, awake subjects is followed by unpredictable responses extending to severe hyperventilation (personal observations), reflecting erratic behavioral responses.

When PAV is applied to normal subjects during steady exercise³¹ or during inhalation of CO_2 -enriched air,²⁸ there is little change in ventilation. Thus, during ventilatory stimulation the assist is primarily utilized to reduce respiratory muscle output rather than to increase ventilation. Georgopoulos et al³² measured P_{mus} during CO_2 rebreathing with and without PAV unloading. They found that P_{mus} was the same when measurements were compared at the same end-tidal P_{CO_2} . Thus, it appears that during exercise-induced or CO_2 -induced ventilatory stimulation the downregulation of P_{mus} , at the same exercise level or the same fraction of inspired carbon dioxide (F_{ICO_2}), is mediated by small reductions in systemic P_{CO_2} .

There are numerous reports on the changes in $\dot{V}_E$ and P_{aCO_2} with PAV in patients with respiratory failure.^{15,16,19–26,33–39} Responses ranged from virtually no change or even a decrease in $\dot{V}_E$ as assist increased^{33–36} to large increases in $\dot{V}_E$ and P_{aCO_2} decreases in.^{16,19,25,37} These differences can be

explained readily if one considers the experimental circumstances of the various studies or patients.

1. *Intubated ventilator-dependent patients with normocapnia.* It is clearly not possible to establish steady-state values of $\dot{V}_E$ and PaCO_2 during unassisted breathing in these patients as they rapidly develop distress. The effect of PAV is, accordingly, determined over a range of assist above a minimum value (e.g., 80% vs. 40%). Furthermore, in such patients, PaCO_2 at the minimum tolerable assist is normal. All such studies demonstrated very little change or even a small decrease in $\dot{V}_E$ as assist increased.^{33–36} PaCO_2 decreased, but the change was small (2 to 4 mm Hg). These patients, therefore, behave like the subject represented by line *a* in Figure 12-4B, who has no nonchemical inputs and a small $\Delta P_{\text{CO}_2, \text{NA-AT}}$. This state, however, is reached only at some finite assist. It therefore would appear that these patients become comfortable only when their PaCO_2 is a few millimeters of mercury above AT. Under these conditions, $\dot{V}_E$ cannot increase further by PAV application (Figure 12-4B, line *a*); the extra assist is used simply to decrease muscle output. The decrease in $\dot{V}_E$ observed in some cases^{34,36} is caused by a reduction in V_D/V_T and/or $\dot{V}_{\text{CO}_2}$ (secondary to decreased respiratory muscle work) because in all such cases, PaCO_2 was lower even though $\dot{V}_E$ was lower.^{34,36}

The preceding observations lead to an interesting conclusion: Ventilator-dependent patients who show little change in $\dot{V}_E$ and PaCO_2 over a relatively wide PAV assist do not tolerate a PaCO_2 that is much above AT. This indicates a high degree of chemosensitivity that likely contributes to their ventilator dependence. *Chemosensitivity*, as used here, does not refer to ventilatory responses (which are affected by mechanics and muscle strength) but to central effects of PaCO_2 on respiratory sensation and muscle activation.

2. *Chronic hypercapnia with and without acute exacerbation.* In virtually all reported studies, $\dot{V}_E$ and PaCO_2 on PAV were compared with values obtained during a period of unassisted breathing.^{16,19–21,23,25,26,40} Unlike the preceding group, there always was a significant increase in $\dot{V}_E$ and PaCO_2 a decrease in. The changes were modest, however (approximately 25% increase in $\dot{V}_E$ and a 3- to 6-mm Hg decrease in PaCO_2). Such a response is consistent with a somewhat larger difference between AT and unassisted P_{CO_2} (subject represented by line *c* in Fig. 12-4B). PaCO_2 remained abnormally high in all cases, consistent with a high AT. The changes in $\dot{V}_E$ and PaCO_2 likely would have been greater if the hypoxic stimulus did not change. In all but one study,¹⁹ PaO_2 was quite low during unassisted breathing and improved with PAV. Because a higher PaO_2 decreases the ventilatory response to CO_2 ,¹³ an increase in PaO_2 mitigates the increase in CO_2 response produced by PAV, resulting in a smaller increase in ventilatory response. The assist provided in this case is used preferentially to decrease muscle output as opposed to increasing $\dot{V}_E$. In one study,¹⁹ the hypoxic drive was negligible at baseline ($\text{PaO}_2 = 99.5$ mm Hg). Here, the increase in $\dot{V}_E$ was much greater (38%¹⁹). Although the changes in PaCO_2 and PaO_2 undoubtedly contributed to the reduction in respiratory muscle output, most of these patients were alert, so a reduction in nonchemical inputs may have been partly responsible.
3. *Acute hypercapnic failure.* Gay et al³⁸ reported an average 8 mm Hg decrease in PaCO_2 (60 to 52 mm Hg) within a half hour of instituting noninvasive PAV. Likewise, in a study by Busterholtz et al on patients with acute cardiogenic pulmonary edema, PaCO_2 decreased from 51 ± 25 to 41 ± 25 mm Hg and PaO_2 increased from 66 ± 18 to 130 ± 30 mm Hg within 30 minutes of applying PAV.⁴¹ Considering that not all patients in these two studies were hypercapnic and that the nonhypercapnic patients likely did not contribute to the average decrease (see 4 below), the decrease in PaCO_2 in the hypercapnic group must have been greater. In another study,³⁷ there were four patients with acute hypercapnic failure not associated with central depression. In them, PaCO_2 declined 17 mm Hg on average (66 to 49 mm Hg) within a half hour of instituting PAV. These observations suggest that patients with acute hypercapnia secondary to severe acute mechanical abnormalities do sustain large increases in $\dot{V}_E$ and reductions in PaCO_2 on institution of PAV (see line *b*, Fig. 12-4C). Interestingly, PaCO_2 remained above normal (approximately 50 mm Hg) in some patients for a few hours.³⁷ It is possible that the apneic threshold increased somewhat during the preceding period of severe hypercapnia.
4. *Acute hypoxemic failure.* Information about this group is scant. Although patients were included in three previous reports,^{37–39} only in one study were the results of the normocapnic group (four patients) separated from those of hypercapnic patients.³⁷ In these four patients, PaCO_2 did not change despite distress decreasing dramatically. The likely explanation for failure PaCO_2 of to decrease is that respiratory muscle output to a large extent was related to nonchemical inputs, which decreased substantially on unloading (pathway 1 in Fig. 12-4D).

Respiratory Rate and Breathing Pattern

With one exception,¹⁶ when there was no clinical distress at the lowest level of assist, application of PAV or further increases in percent assist did not result in appreciable changes in ventilator rate (i.e., >2 to 3 breaths/min). This applied to normal sleeping subjects,^{6,17} intensive care unit (ICU) patients in whom percent assist was changed over a wide range above a comfortable level,^{33–36} and ambulatory patients with chronic respiratory failure in whom the lowest level was spontaneous breathing.^{19,20,24–26,40} In the only exception,¹⁶ respiratory rate (RR) decreased substantially when PAV was applied, but in this case there were clear signs of runaway (i.e., patient was no longer in the PAV mode).

In patients with acute exacerbation of chronic obstructive pulmonary disease (COPD), RR on PAV was lower than during spontaneous breathing (3 to 5 breaths/min^{21,23}). The pH, however, was low at baseline, and some degree of distress may have been present then. When PAV is applied to patients with clear respiratory distress, RR decreases dramatically along with relief of distress.^{37–39}

From these observations it is clear that PAV does not per se change RR. RR changes only when PAV relieves respiratory distress. In physiologic studies in which respiratory drive is deliberately increased, RR does not increase until moderate levels of stimulation.^{18,42} This applies whether stimulation is produced by hypercapnia, hypoxemia, acidosis, or an increase in metabolic rate.⁴³ Thus, a change in RR with assist level indicates that respiratory drive is in a range where RR is sensitive to drive and hence probably excessive, whereas failure of RR to change over a range of PAV assist indicates that respiratory drive is only modest over this range. There are two important clinical implications to these observations on PAV:

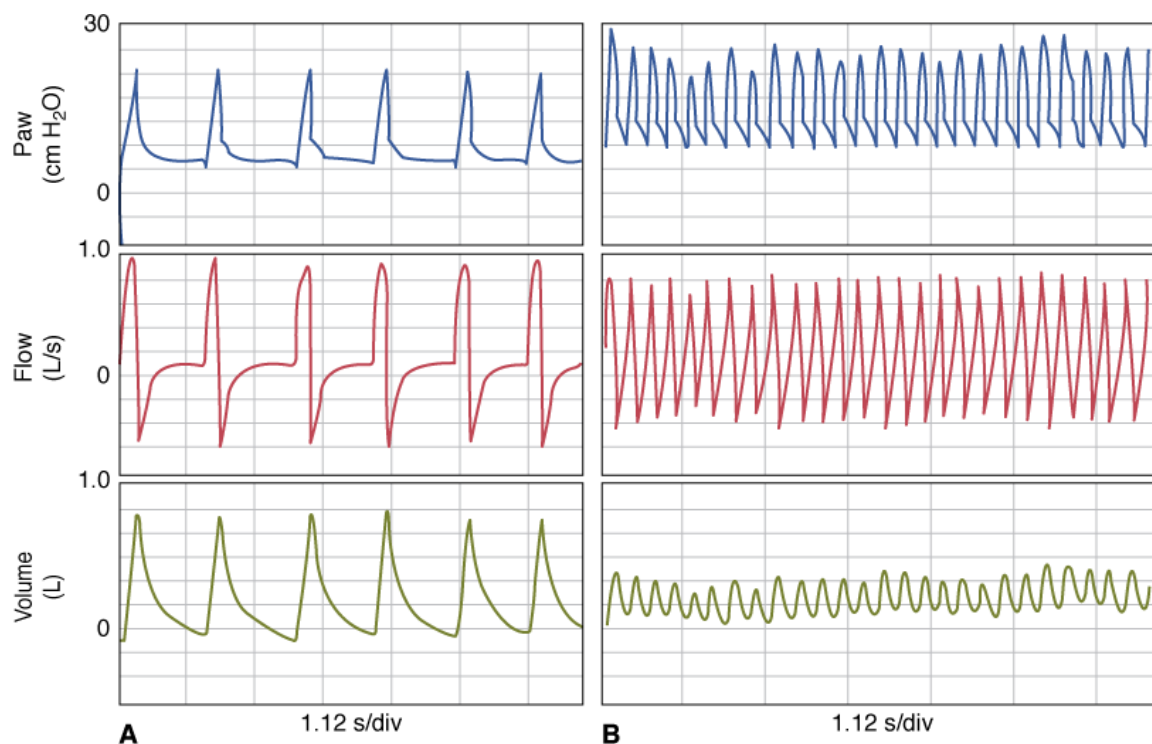
1. Failure of RR to change over a range of PAV assist indicates that respiratory drive is only modest over this range and that the RR observed in this range is the undistressed value preferred by the patient's control system. Importantly, undistressed RR, so defined, ranges from 12 to 46 breaths/min.^{33,34} That undistressed RR varies widely among patients is consistent with the wide range in normal subjects (8 to 25 breaths/min).⁴⁴ The main difference between ICU patients and normal subjects is that the average undistressed rate is 10 breaths/min higher.³³ A number of factors may contribute to this, including body temperature, irritation of tracheal receptors by the endotracheal tube, disease-related effects on pulmonary and other receptors, neuropathology, and drug effects.³³
2. Because the undistressed RR can be quite low, a change in RR as assist level is changed is more important than absolute RR at the low assist. For example, an increase from 20 to 25 breaths/min may indicate distress, although 25 breaths/min is not usually considered a sign of distress. By contrast, RR in excess of the usual cutoff of 35 breaths/min need not reflect distress.

Tidal volume responses mirror the $\dot{V}_E$ responses and obviously share the same mechanisms (see “Ventilation and Partial Pressure of Carbon Dioxide” above). It is important to note that in normocapnic patients, once a distress-free assist level is reached, further increases have little effect on V_T .^{33,34}

Accordingly, every patient has a preferred or target V_T that cannot be exceeded with more PAV assist. As with normal subjects,⁴⁴ the preferred V_T varies widely among patients (4 to 15 mL/kg^{33,34}).

PAV has made it possible to determine the undistressed breathing pattern in ICU patients. This proved quite variable. Figure 12-5 shows two extremes. With this wide range, a one-size-fits-all strategy of mechanical ventilation (e.g., setting a target V_T) is clearly not ideal (see “Physiologic Consequences of Operational Differences” below).

Figure 12-5



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Tracings from two patients on high PAV assist showing extremes of undistressed breathing pattern. *Paw*, airway pressure.

Large breath-by-breath variability in V_T is characteristic of normal breathing, particularly in wakefulness.^{45,46} Variability decreases in patients with abnormal mechanics.^{47,48} Probably because PAV improves neuroventilatory coupling, breath-by-breath variability tends to be large in this mode (see, e.g., Fig. 12-1). Coefficients of variation of 25% or more are not unusual,^{23,25,35-37} and spontaneous sighs may be frequent. As with normal subjects,⁴⁹ breathing variability on PAV is less in sleeping and obtunded patients (personal observations).

Ventilatory Instability

The tendency for the respiratory system to become unstable is described by the so-called loop gain.⁵⁰⁻⁵² A value of 1 indicates that recurrent cycling will occur spontaneously (e.g., Cheyne-Stokes breathing). The lower the value, the more stable is the system. Ventilatory response to chemical stimuli is an important determinant of loop gain.⁵⁰⁻⁵² Because PAV increases ventilatory responses, we were concerned initially that it might precipitate periodic breathing.¹ This, however, did not materialize. Although PAV may aggravate preexisting Cheyne-Stokes breathing,⁵³ there are no reports of PAV-induced Cheyne-Stokes breathing in the usual ICU patient, and we have observed only a few in whom breathing became periodic on high PAV support.

The resistance to Cheyne-Stokes breathing was explained recently. Normal subjects require threefold to fourfold amplification of ventilatory responses to develop periodic breathing.^{6,17,54} Thus, when respiratory muscles and mechanics are normal, native loop gain is less than 0.3. In the average ICU patient, respiratory muscle strength is 50% of normal,⁵⁵ resistance is four times normal (14 vs. 3 to 4 cm H₂O/L/s⁵⁶), and elastance is two to three times normal (28 vs. 10 to 14 cm H₂O/L⁵⁷). Collectively, these abnormalities should decrease ventilatory responses to 20% of the normal value. For PAV to induce periodic breathing in the average patient, it first must normalize ventilatory responses (i.e., a fivefold increase in the average patient) and then three to four times more, a greater than 10-fold amplification. This is very difficult to achieve because of technical and physiologic limitations (see “Limitations” below). Accordingly, if periodic breathing develops on PAV, it suggests that (a) respiratory muscles and mechanics are near normal, and the patient likely does not need ventilator support, and/or (b) the chemical control system is inherently unstable, and one should suspect disorders that result in Cheyne-Stokes breathing, chiefly heart failure. A third condition that may precipitate periodic breathing is the runaway phenomenon (see below). Here, large tidal volumes may result, precipitating hypocapnia and recurrent central apneas. The pattern is unlike the crescendo-decrescendo Cheyne-Stokes breathing variety, however, and more like that produced by pressure support (several large breaths alternating with apnea^{6,58}).

The ability of PAV to increase loop gain by measurable quantities is currently being used to study mechanisms of instability during sleep.^{52,59}

Responses during Exercise

Application of PAV during submaximal exercise in patients with severe COPD increased endurance time and reduced the rate of progression of dyspnea.^{29,60–62} Patients with very severe COPD who received PAV during exercise in a rehabilitation program demonstrated greater improvement in unassisted exercise tolerance relative to a control group.⁶³ A beneficial effect of assist was not evident in mild COPD.⁶⁴ Application of PAV during submaximal exercise also increased endurance time and reduced dyspnea in patients with idiopathic pulmonary fibrosis⁶⁵ and in obese patients.⁶⁶

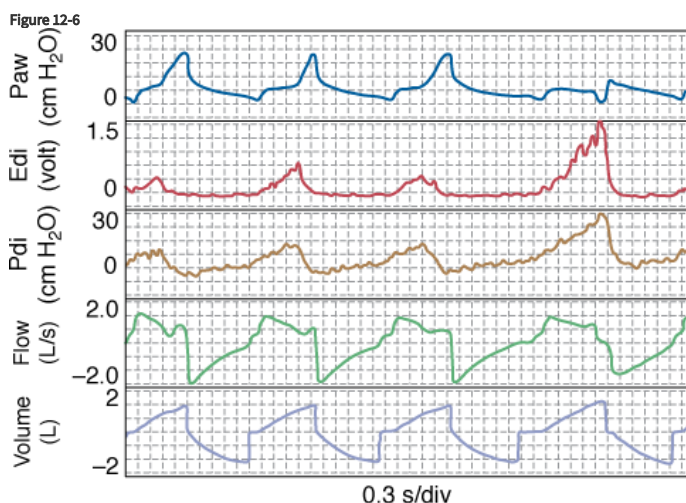
Apart from its potential therapeutic role, PAV also has been used to examine the role of respiratory muscles in limiting exercise in normal subjects and in patients with COPD, with highly interesting results.^{31,67–74}

Comparison with Other Modes

Operational Differences between Proportional-Assist Ventilation and Other Modes

With PAV, the assist (i.e., P_{aw}) varies directly with the intensity of patient effort (see Fig. 12-1). By contrast, with pressure-support ventilation (PSV), the assist is the same breath after breath regardless of intensity of effort. With volume-controlled ventilation (VCV), assist varies inversely with effort (Fig. 12-6). This is so because flow and volume are preset. If the patient's contribution increases, the ventilator must deliver less assist (P_{aw}), and vice versa.

Otherwise, delivered flow and volume will deviate from set values. These different relations have been well documented.^{18,22,75}



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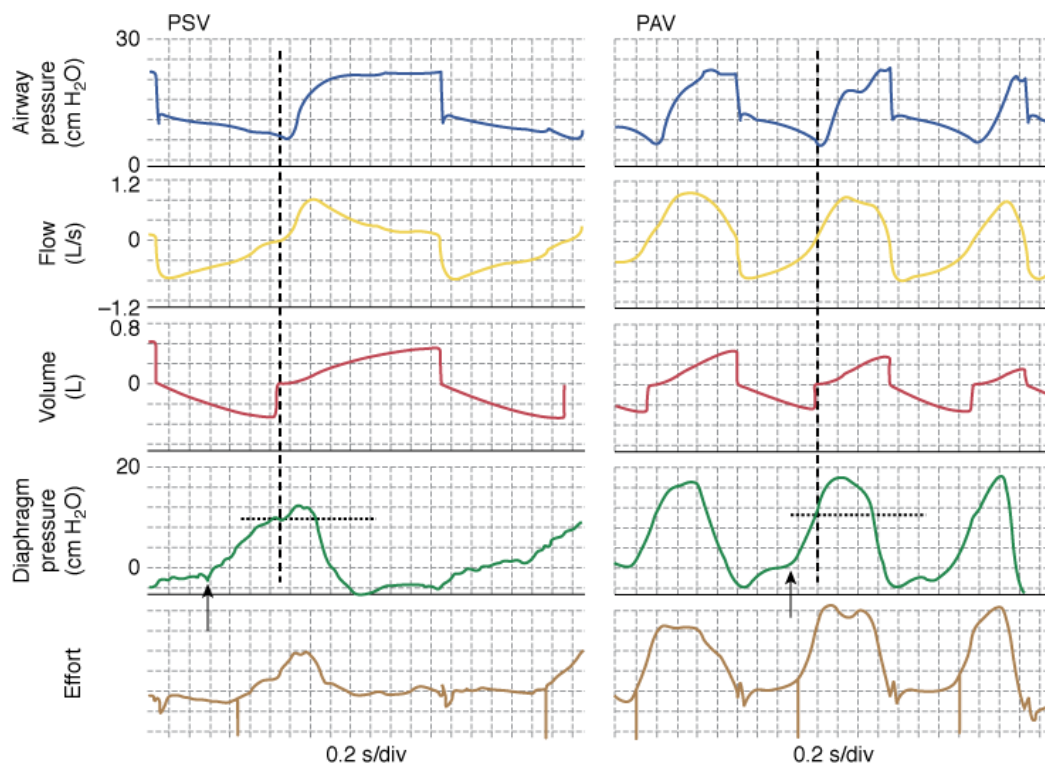
Tracings from a patient on volume-controlled ventilation. Note that ventilator cycles extend beyond inspiratory effort in the first three breaths while terminating during the effort in the last breath. Note also that patient received little or no assist when his effort was greatest (last breath). *Edi*, diaphragmatic activity; *Paw*, airway pressure; *Pdi*, diaphragmatic pressure.

With PAV, the end of the ventilator cycle is synchronized automatically with the end of patient effort (see Fig. 12-1), whereas with other modes it is not.

Although ventilator response delays tend to delay cycling off somewhat,⁷⁶ the effect is fairly trivial compared with the situation in other modes (see “Ventilator Response Time” below). In VCV, there is no relationship whatsoever; the patient determines the end of his effort, whereas the caregiver determines the end of the ventilator's cycle. Any synchrony is happenstance. The ventilator may continue inflation well after the end of effort, when patient wants to exhale (e.g., breath 1 in Fig. 12-6), or may cycle off, withdrawing support before the end of effort while patient is still trying to inhale (e.g., breath 4 in Fig. 12-6).

With PSV, synchrony between the ends of the ventilator's and the patient's inspiratory phases may or may not occur depending on the patient's respiratory mechanics and the relation between PSV level and P_{mus} .^{77,78} Ventilator cycles often extend well beyond inspiratory effort^{34,79} or may be almost completely out of phase with them⁷⁹ (Fig. 12-7A). At times, inflation extends over two or more efforts (Fig. 12-8A).³⁴

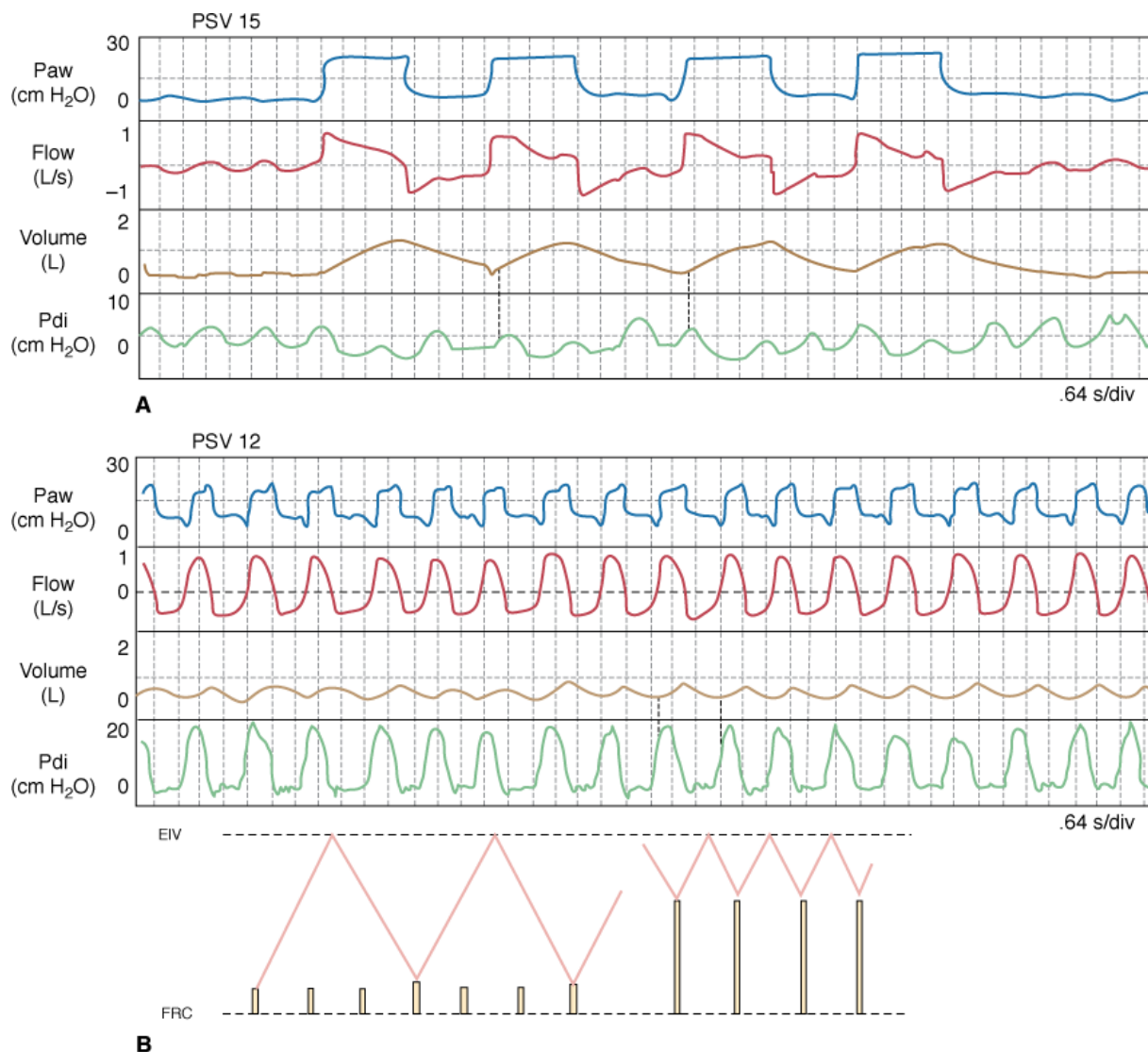
Figure 12-7



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Comparison of pressure-support ventilation (PSV) and proportional-assist ventilation (PAV) in a patient with severe dynamic hyperinflation. With both modes, inspiratory muscles had to generate 10 cm H₂O before inspiratory flow could be generated and the ventilator triggered (*vertical dashed lines*). Note that in PSV, ventilator cycle extends well into neural expiration. There also were many ineffective efforts (not shown). This is not the case in PAV. Note also that diaphragmatic output is considerably higher in PAV. As a result, delay between onset of effort (*arrows*) and triggering is much less. In addition, respiratory rate is higher with PAV. With severe dynamic hyperinflation, it is difficult to maintain respiratory muscle output at a low level in PAV. See [Figure 12-13](#) for transition from PSV to PAV in this patient. *Bottom tracing*. A semiquantitative effort signal generated without knowledge of respiratory mechanics¹¹⁷ that can be used to identify onset and end of efforts noninvasively in real-time (*event marks*). Note that the onset of effort (*downgoing marks*) can be identified well before inspiratory flow crossing. If used for triggering, this essentially can eliminate the extra work associated with dynamic hyperinflation.

Figure 12-8



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A dramatic example of a marked shift from slow, deep breathing (panel A) to very rapid, shallow breathing (panel B) following a small reduction in pressure support (PSV). The increase in respiratory rate was artifactual and related to improved synchrony (note that the rate of diaphragmatic efforts was unchanged). The small tidal volumes are related to dynamic hyperinflation (note that Pdi increases substantially before flow becomes inspiratory) (*vertical dotted lines*). Inset shows schematically the mechanism of shallow breathing. Height of *solid lines* represents effort amplitude. See text for additional details. EIV, end-inspiratory volume; FRC, functional residual capacity; Paw, airway pressure; Pdi, diaphragmatic pressure.

The operational characteristics of NAVA⁸⁰⁻⁸² are very similar to those of PAV in that the assist is proportional to instantaneous effort in both cases. Thus, in both cases the assist will increase when effort increases and vice versa, and in both cases the ventilator cycle will terminate soon after the end of inspiratory effort. The main difference is in the signal used to drive the ventilator: PAV uses an indirect noninvasive estimate of effort (calculated Pmus) whereas NAVA uses a direct estimate of diaphragmatic activity obtained from internally inserted esophageal electrodes. This gives NAVA an advantage with respect triggering in that with NAVA the assist can, theoretically, begin very soon after the onset of diaphragmatic activity in the presence of important dynamic hyperinflation or circuit leaks, whereas with PAV the assist will not start until flow becomes inspiratory, and triggering and assist may be adversely affected in the presence of significant leaks (see “[Limitations](#)” below). On the other hand, because the diaphragm participates in many nonrespiratory activities (such as postural changes and vomiting), control of airway pressure by diaphragmatic activity could result in undesirable large increases in airway pressure during such activities. For example, the diaphragm contracts vigorously during vomiting to increase abdominal pressure. This may be expected to cause a large increase in airway pressure in the case of NAVA. PAV, in contrast, is not susceptible to this problem because diaphragmatic descent and reduction in intrathoracic pressure are prevented by concurrent contraction of the expiratory muscles.

Physiologic Consequences of Operational Differences

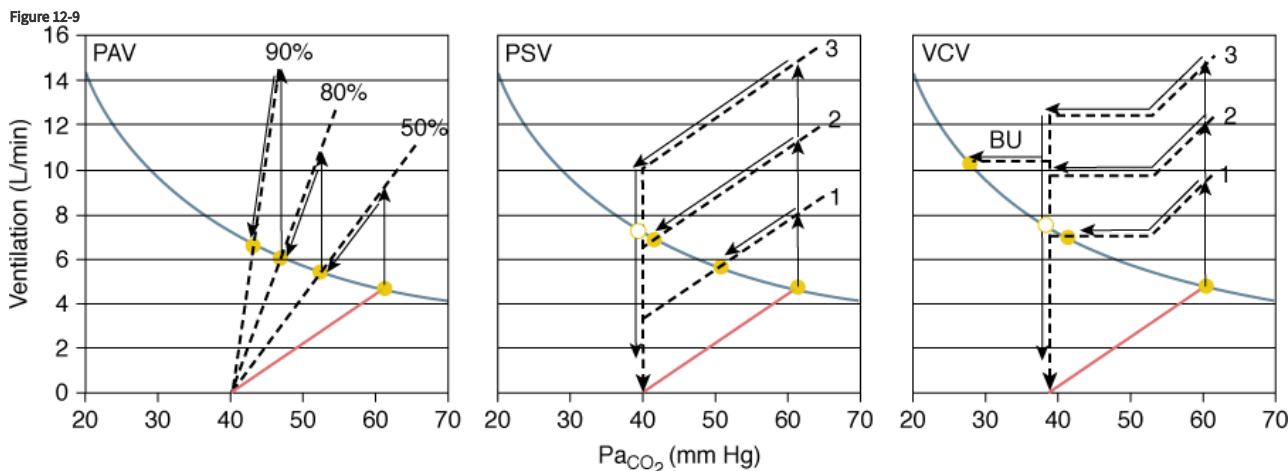
The following sections deal with consequences as they relate to conventional assist modes (PSV and VCV). Because NAVA has similar operational characteristics, its consequences vis-a-vis conventional assist modes are expected to be similar (see [Chapter 13](#) and reference 82 of this chapter). There are

no studies in which NAVA and PAV have been compared directly so as to ascertain the clinical impact of NAVA's triggering advantage in the presence of dynamic hyperinflation and leaks.

Response to Different Levels of Assist

Figure 12-9 illustrates what, theoretically, should happen when assist level is varied in different modes (see refs. 3 and 83 for more details). The response of RR to changes in PaCO_2 near the AT is key to understanding these plots. As indicated earlier, RR is fairly constant over a range of PaCO_2 above the AT. For

example, average difference between spontaneous RR and RR just before apnea is less than 1 breath per minute.⁶ At AT, breathing simply stops. There is no gradual reduction in RR.



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Responses to different levels of assist with proportional-assist (PAV), pressure-support (PSV), and volume-controlled ventilation (VCV). The ventilatory response to CO_2 is shown for three assist levels in each case. The format is as in Figure 12-4. Note that in PAV the ventilatory response line intersects the metabolic hyperbola at all levels of assist. A steady state thus is possible at all levels. In PSV and VCV, the response lines intersect the hyperbola over a limited range of assist. At higher levels, a steady state is not possible, and nonsynchrony must result (see text). In this simulation, respiratory rate is shown to increase when PaCO_2 exceeds 53 mm Hg. This accounts for the increase in ventilation above this PaCO_2 in VCV. BU, backup rate.

The effect of different levels of PAV (see Fig. 12-9A) was discussed earlier (see Fig. 12-4). A steady state is possible at all assist levels, except in rare cases where chemical control is highly unstable (see “Ventilatory Instability” above). Once the AT is approached, it is not possible to increase $\dot{V}_E$ further.

The relationship between P_{CO_2} and ventilation during PSV is extremely complex.^{3,34,77,83} For simplicity, it is shown as a parallel shift in the ventilatory response, reflecting the fact that the assist is independent of P_{CO_2} . Exactly what happens, however, to the slope of the ventilatory response is not terribly important here. What is important is that with PSV a minimum $\dot{V}_T$ is delivered once the ventilator is triggered. This minimum $\dot{V}_T$ is a function of PSV level and respiratory mechanics.^{6,77} For example, in a patient with an elastance of 25 cm $\text{H}_2\text{O}/\text{L}$ and a resistance of 12 cm $\text{H}_2\text{O}/\text{L}/\text{s}$, the minimum $\dot{V}_T$ at PSV of 7, 14, and 20 cm H_2O will be approximately 0.22, 0.43, and 0.64 L, respectively.^{6,77} If the patient's RR near the AT is a conservative 16 breaths/min, and every effort triggers the ventilator, minimum $\dot{V}_E$ near the AT will be 3.5, 6.9, and 10.2 L/min for the three levels, respectively (see Fig. 12-9). A higher minimum RR (range: 12 to 46 breaths/min; see “Respiratory Rate and Breathing Pattern” above) would increase minimum $\dot{V}_E$ correspondingly.

For the patient in Figure 12-9, applying PSV at level 1 will boost ventilation initially above the metabolic hyperbola. PaCO_2 falls along the response line of that level. Because minimum is below the hyperbola (left end of line 1), a steady state can be reached (solid dot on line 1). For level 2, a steady state still can be reached, but PaCO_2 will be just above the AT, and efforts will be very feeble. At level 3, a steady state is not possible because minimum $\dot{V}_E$ is above the metabolic hyperbola. When PaCO_2 is above the AT, $\dot{V}_E$ is above the hyperbola, and P_{CO_2} must fall. When it falls below the AT, $\dot{V}_E$ becomes zero. The only steady-state $\dot{V}_E$ is the open circle. This is not possible, however, if all efforts trigger the ventilator. For average $\dot{V}_E$ to equal $\dot{V}_E$ at the open circle, ventilator rate must decrease below the patient's minimum RR. This occurs in one of two ways depending on the time constant of the respiratory system (resistance/elastance [R/E]). If R/E is very short (e.g., severe, restrictive disease), ventilator cycles will not encroach on neural expiration,^{77,78} allowing lung volume to return to FRC before the next effort. Here it is possible to continue triggering until the AT, and recurrent central apneas develop. When R/E is long, the ventilator cycle is also long^{77,78} and extends into neural expiration. As efforts weaken, more and more efforts fail to trigger the ventilator. Here,

ineffective efforts are scattered between triggered breaths. Depending on R/E, ineffective efforts may appear well before the AT. Because R/E is more commonly long than short, intermittent ineffective efforts are more common than recurrent central apneas.^{34,84}

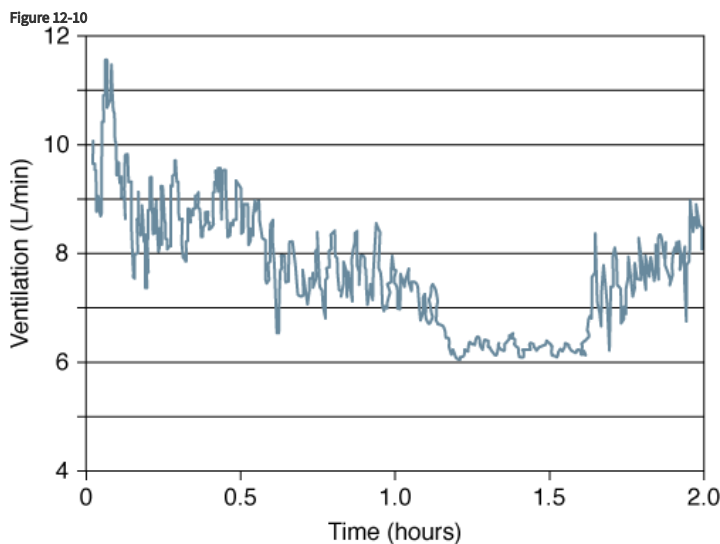
In summary, with neither PAV nor PSV it is possible to decrease average PaCO_2 below the AT. What is different is that with PAV, V_T is independent of assist level near the AT, and ventilator cycles cannot extend substantially into neural expiration. Thus, a steady state can be reached at all levels without nonsynchrony.³⁴ With PSV, by contrast, V_T increases monotonically with assist. At some level the product $V_T \times \text{minimum RR}$ exceeds the $\dot{V}_E$ required for steady state, and nonsynchrony must occur. These differences have been demonstrated experimentally.³⁴

With VCV, $\dot{V}_E$ is constant over the range where RR is independent of drive and is given by patient RR $\times$ set V_T . As set V_T increases, $\dot{V}_E$ increases and remains at that level as efforts decline (*horizontal lines, right panel, Fig. 12-9*). If steady-state PaCO_2 associated with this fixed $\dot{V}_E$ is above the AT, a steady state with maintained synchrony can result (e.g., *line 1*). If not (*lines 2 and 3*), and there is no or minimal backup rate, ineffective efforts or recurrent central apneas must result (as with PSV), and nonsynchrony will increase as set $\dot{V}_T$ increases.⁸⁴ With a backup rate, once the AT is reached, $\dot{V}_E$ will decrease to set $V_T \times \text{backup rate}$. If this $\dot{V}_E$ is below the hyperbola at the AT, periods in which the ventilator is triggered by the patient will alternate with periods in which it is self-triggered as PaCO_2 oscillates about the AT. If backup $\dot{V}_E$ is above the hyperbola (*line BU in Fig. 12-9*), PaCO_2 must fall further. Depending on ventilator settings, marked hypocapnia may develop.

In summary, with PSV and VCV there is a limited range of assist that is consistent with synchrony. Above this range, ineffective efforts or recurrent central apneas must develop. It is evident that the “appropriate” assist range depends on the metabolic hyperbola (a function of metabolic rate and V_D/V_T) and minimum patient RR near the AT (see *Fig. 12-9*). Because these may change with time, a suitable level may become unsuitable at another time. Several studies confirm that nonsynchrony is quite common with PSV and VCV^{24,34,79,84} while being virtually nonexistent at all levels of PAV.^{19,24,34} Ineffective efforts do occur at times during PAV, but they are very infrequent³⁴ and usually occur when a large breath (spontaneous sigh or runaway breath) is followed by a weak effort.

Response to Changes in Ventilatory Demand

Figure 12-10 is a representative example of spontaneous changes in $\dot{V}_E$ over a 2-hour period. The changes are not trivial; the highest level was almost twice the lowest. This is not surprising because many influences that affect ventilatory demand can change in these patients, for example, pain, anxiety, sleep-wake cycles, metabolic rate, pH, and drugs. It is therefore useful to examine what happens with the three modes in response to such changes.

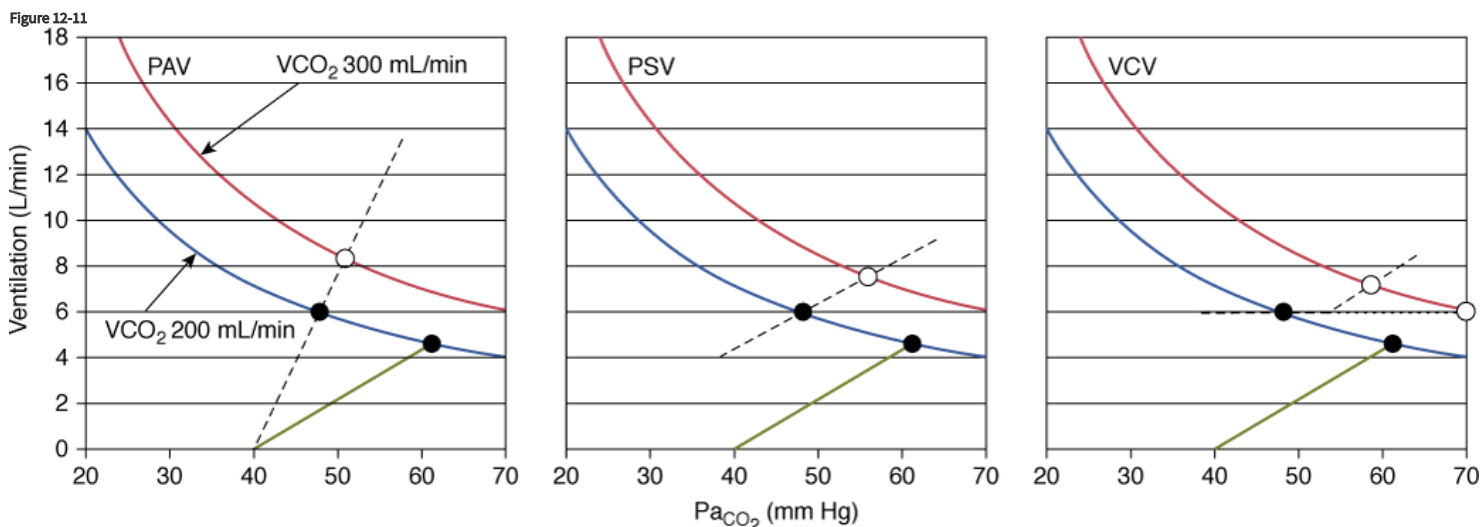


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One-minute moving average of ventilation over a 2-hour period on PAV. Note large spontaneous changes in ventilatory demand.

Figure 12-11 illustrates two metabolic hyperbolas, representing two metabolic rates. The unassisted ventilatory response (*solid line*) is identical and would be associated with a PaCO_2 of 62 mm Hg. Initial ventilator settings with the three modes were adjusted to produce identical $\dot{V}_E$ (6 L/min) and PaCO_2 (48 mm Hg) at the lower metabolic rate. Ventilatory demand increases by 50% (upper hyperbola). Efforts increase, but RR initially does not. With PAV, assist increases, and $\dot{V}_E$ increases along a relatively steep response line, reaching the higher hyperbola at a PaCO_2 of 51 mm Hg. With PSV, assist is constant. As a

result, the ventilatory response is no better than the unassisted slope. The higher hyperbola is reached at 57 mm Hg. With VCV, ventilation cannot increase until RR increases. Without tachypnea, the metabolic hyperbola is reached at a PaCO_2 of 70 mm Hg.



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Responses to a 50% increase in ventilatory demand with proportional-assist (PAV), pressure-support (PSV), and volume-controlled ventilation (VCV). The unassisted ventilatory response to CO_2 is given by the solid diagonal line in each panel. The three modes were set during the low-demand period to produce the same ventilation and PaCO_2 (solid circles/dashed lines). Because of the higher ventilatory-response slope on PAV, the new ventilation could be reached with a much smaller increase in PaCO_2 (open circles). See text for more details.

It is clear that the likelihood of respiratory distress and tachypnea developing is higher with VCV and PSV. With VCV, there is the added problem that ventilator inspiratory time (T_I) does not change as patient respiratory cycle time (T_{TOT}) decreases. Less time remains for exhalation, promoting nonsynchrony.⁸⁴ Nonsynchrony occurring at a time of high respiratory drive may trigger anxiety. Of course, both PSV and VCV can be readjusted to provide adequate support at the higher demand. This level, however, will be excessive when ventilatory demand returns to the lower level.

The different responses to CO_2 challenge under the three modes were well illustrated in normal subjects.¹⁸ In ventilator-dependent patients, Ranieri et al⁷⁵ showed that following addition of dead space during PAV, V_T increased with no change in RR (20.1 to 19.8 breaths/min). When added during PSV in the same patients, there was little change in V_T , and RR increased dramatically (from 16.4 to 33.2 breaths/min). By contrast, a recent repetition of this study⁸⁵ failed to demonstrate differences between PSV and PAV in the ventilatory, pressure-time product or transdiaphragmatic pressure responses to added dead space even though the magnitude of assist increased with PAV and not with PSV. There are several reasons to explain this negative result. Although both studies added the same amount of dead space (150 mL), the challenge was much greater in the Ranieri⁷⁵ study, as evidenced by a much greater increase in V_E (approximately equal to 13.5 vs. 2.1 L/min), PaCO_2 (approximately equal to 19.0 vs. 2.0 mm Hg), and pressure-time product per minute (185 vs. 32.4 cm $\text{H}_2\text{O}\cdot\text{s}/\text{min}$) on PSV of 10 cm H_2O . The reason for the large difference in challenge presented by the same addition of dead space is not clear. Nonetheless, with a minimal challenge, as was delivered in the study of Varelmann et al,⁸⁵ it is difficult to obtain significant responses given the noise in such studies (spontaneous changes in respiratory drive). It is worth noting that in the latter study,⁸⁵ PaCO_2 increased more when dead space was added with PAV (5 mm Hg on PAV vs. 2 mm Hg on PSV) even though V_T and V_E increased more during PAV, albeit not significantly so. Thus, ventilatory demand was greater during added dead space on PAV than when it was added during PSV. Also, as pointed out by Mols et al⁸⁶ the two studies used different ventilators (Winnipeg Ventilator vs. Dräger Evita 4) with very different response characteristics.

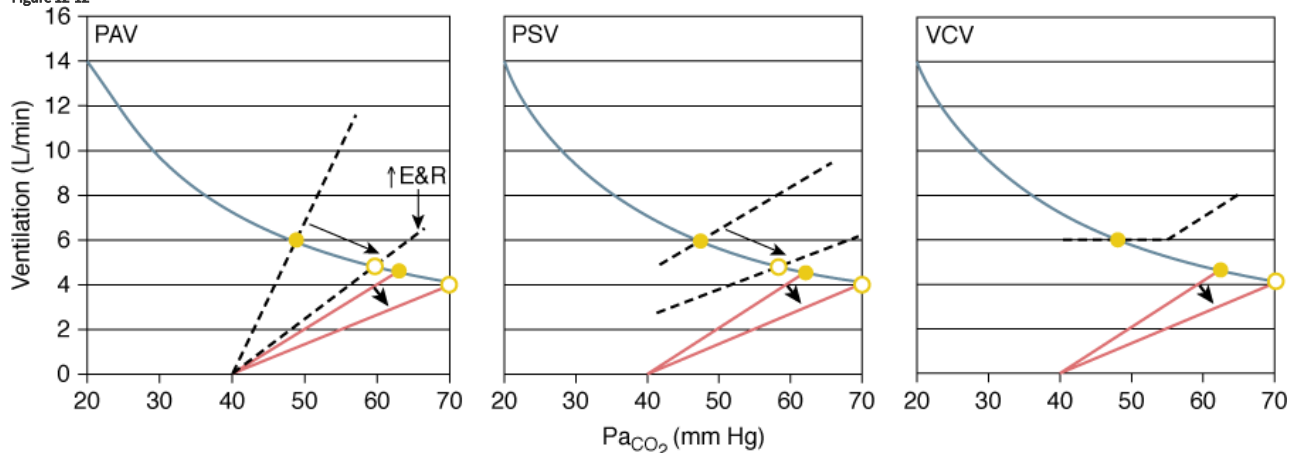
Muscular exercise represents the most extreme form of ventilatory stimulation. The impact of PAV and PSV on exercise endurance and dyspnea in COPD patients was compared in one study.⁶¹ PAV was found superior.

Response to Changes in Respiratory Mechanics

Substantial changes in resistance (R) may occur from time to time.⁵⁶ Although similar information about elastance (E) is not available, there is every expectation that it also changes from time to time (e.g., secondary to changes in lung water, abdominal pressure, or atelectasis). Accordingly, it is important to consider the response to changes in mechanics.

Figure 12-12 illustrates the effect of a combined 50% increase in R and E. At a given P_{mus} , ventilation is inversely related to mechanical properties. Thus, a 50% increase in E and R reduces the unassisted ventilatory response to CO_2 to 67% of baseline.

Figure 12-12



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Responses to a 50% increase in elastance and resistance with proportional-assist (PAV), pressure-support (PSV), and volume-controlled ventilation (VCV). *Solid diagonal lines*: Unassisted ventilatory responses before and after the change in mechanics. *Dashed lines*: Ventilatory responses on assisted ventilation. All modes were set before the change to produce the same ventilation and Pa_{CO_2} (solid circles/dashed lines). As mechanics worsen, Pa_{CO_2} increases and ventilation decreases with PAV and PSV (open circles) but not with VCV.

An increase in R and E will reduce the slope of the ventilatory response under PAV for two reasons. First, percent assist is now lower. For example, if $\dot{V}_A$ were 70% E, and E increased by 50%, $\dot{V}_A$ would become 47% (70/150). This reduces the amplification factor (from 3.3 to 1.88 in this case; amplification factor = $100/[100 - \text{percent assist}]$). Second, the slope of the ventilatory response that is being amplified also has decreased to 67% of the initial value. As a result, ventilatory response on PAV is now 38% of its initial value. $\dot{V}_E$ must fall, Pa_{CO_2} must rise, and distress may develop.

The situation is comparable with PSV. The already low ventilatory slope becomes even lower because the unassisted slope is now lower. Furthermore, the same pressure assist will be less effective in boosting $\dot{V}_T$ because of worse mechanics. As a result, the ventilatory response is displaced downward as well (e.g., Fig. 4 in ref. 22). A new steady state is reached at a lower $\dot{V}_E$ and higher Pa_{CO_2} . The changes in $\dot{V}_E$ and Pa_{CO_2} are comparable in the two modes.

By contrast, with VCV, a change in E and R will have no effect on ventilation. The ventilator will deliver the same $\dot{V}_T$. Because there is no change in Pa_{CO_2} , there is no increase in effort. In this respect, therefore, VCV is superior to both PAV and PSV.

Grasso et al²² compared responses to an increase in elastance with PAV and PSV. Surprisingly, they found that with PAV, $\dot{V}_T$ decreased less, RR increased less, and the increase in dyspnea was less. It is not clear why this was so. Regardless, the better results in this study were not expected and should not be viewed as intrinsic to PAV.

An improvement in mechanics also can create problems with PAV and PSV but not with VCV. If percent assist is high before the change, a reduction in R or E may cause the percent assist to exceed 100% of the new R or E, resulting in runaway. With PSV, improvement in mechanics at the same assist level will increase $\dot{V}_E$ and Pa_{CO_2} decrease. Asynchrony may appear.

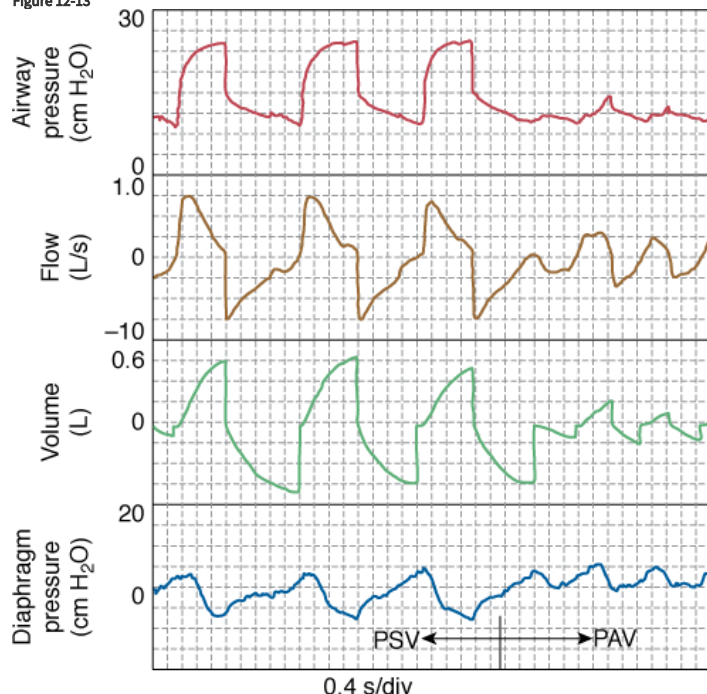
In summary, on standard PAV, changes in mechanics may be followed by distress or excessive ventilator alarming (runaways). This emphasizes the need for using PAV systems equipped with algorithms that continuously monitor respiratory mechanics. This problem is mitigated by the fact that monitoring passive R and E continuously is possible in the PAV mode (see “Noninvasive Monitoring of Respiratory Mechanics, P_{mus} , and Work of Breathing” and “Commercially Available Proportional-Assist Ventilation Delivery Systems” below). Kondili et al⁸⁷ studied the response to changes in respiratory mechanics in ventilator-dependent patients using such a PAV delivery system (PAV+, Covidien). They found that the increase in all indices of respiratory effort was significantly less when the load was added during PAV+ than when added during PSV.

Respiratory Rate in Different Modes

With all modes, patient RR will increase if assist is inadequate. Patient RR, however, is also sensitive to reflexes, independent of chemical drive. Continued inflation during neural expiration prolongs neural expiration and, by extension, reduces RR.^{88–90} This reflex is less pronounced in ICU patients⁹⁰ than in alert patients^{88,89} but is still quite evident. Its gain varies widely among patients.⁹⁰ Figure 12-8 shows a very weak response. Note that patient RR was the

same whether efforts occurred during inflation or deflation. In contrast, [Figure 12-13](#) illustrates a strong response. Here, on PSV, ventilator cycle extended well into neural expiratory time. When PSV was discontinued, there was a marked increase in patient RR (from 20 to 33 breaths/min). This cannot represent distress because it occurred immediately, and diaphragmatic swings were still very low (approximately 5 cm H₂O). Thus, with PSV, nonsynchrony may reduce patient RR for reasons that have little to do with relieving distress. As a corollary, reduction in asynchrony, as would occur if PSV levels were reduced,^{34,77} may result in acceleration of patient RR that is unrelated to distress. This is not a problem with PAV because the ends of patient and ventilator inspiratory phases are synchronized. Therefore, changes in RR on PAV reflect level of distress more reliably.

Figure 12-13



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Transition from pressure-support ventilation (PSV) to PAV in the same patient as in [Figure 12-7](#) with marked dynamic hyperinflation. Efforts were quite weak on PSV and barely succeeded in triggering the ventilator. On switching to PAV, the patient receives very little assist because, unlike PSV, assist cannot outlast inspiratory effort. Effort must increase in order to advance triggering and obtain adequate ventilation (see [Fig. 12-7B](#)). Note that respiratory rate increased suddenly on the switch. Because efforts were still quite weak and the response was immediate, the increase in rate was not caused by a high respiratory drive (i.e., distress) and almost certainly was reflexive in origin secondary to removal of inflation during neural expiration (see text).

Tidal Volume in Different Modes

As indicated earlier (under “[Respiratory Rate and Breathing Pattern](#)”), with PAV, V_T is determined by the patient, and the preferred V_T ranges from 4 to 15 mL/kg, with an average of 7 mL/kg.^{33,34} With VCV, V_T is set without knowledge of the patient’s preferred V_T . PSV is also usually adjusted to yield a given V_T . In either case, the set V_T almost always will be larger than necessary. For example, if V_T is set to 10 mL/kg, it will be greater than preferred V_T in most patients. In the others, it will be inadequate. If one individualizes the assist to comfort level, V_T still will be higher than with PAV for two reasons: first, when V_T is titrated to the lowest level consistent with comfort, the chosen V_T is greater than the average V_T during spontaneous breathing.^{8,91} Second, if V_T or PSV level is set to comfort level at a given point, a change in ventilatory demand (see, e.g., [Figs. 12-10](#) and [12-11](#)) will cause it to become either excessive or too little later. If it becomes excessive, it will not be adjusted down. If it becomes inadequate (increase in demand), however, it will be increased and remain at the high level after, when demand decreases again. On average, V_T will be larger than if it were allowed to vary with demand (i.e., PAV).

Another peculiarity unique to PSV is that dynamic hyperinflation may cause tidal volume to be inversely related to inspiratory effort (see, e.g., [Fig. 12-8](#)). This is so because at a given PSV, the end-inspiratory volume, relative to passive FRC, at which the ventilator cycles off, in the absence of effort (V_{th}), is fixed.⁷⁷ A stronger effort will trigger the ventilator at a higher volume relative to FRC, leaving less difference between the onset of the breath and V_{th} (see inset of [Fig. 12-8](#)). With high PSV, it is possible to reduce chemical drive to very near the AT, resulting in very weak efforts (see, e.g., [Fig. 12-8A](#)). Triggering then can occur only when volume is very close to FRC. When a breath is triggered, volume must increase by a large amount before the ventilator cycles off. Large protracted V_T s result (see [Fig. 12-8A](#)). If efforts increase, either spontaneously or after reducing PSV level, breaths can be triggered at a higher volume and hence closer to V_{th} . A small V_T will cause V_{th} to be reached. Once effort ceases, flow decreases to the cycling-off threshold, and the ventilator cycle

ends. Synchrony is reestablished, but breathing becomes faster and shallower (see [Fig. 12-8](#)). Thus, paradoxically, under these conditions V_T is largest when effort is weakest, and changes in V_T no longer reflect effort.

This scenario can never happen with PAV in part because in the presence of dynamic hyperinflation it is not possible to reduce efforts to the very low levels that can be reached with high-level PSV (see “[Dynamic Hyperinflation](#)” below) and in part because the inflation cycle ends when patient effort ceases regardless of the end-inspiratory volume reached.

Comfort

Comfort, of course, cannot be compared in obtunded patients. Whenever PSV and PAV were compared in alert patients with respiratory distress, however, PAV was preferred.^{23,38,39}

Hemodynamic Effects

Mechanical ventilation has complex effects on the circulation (see [Chapter 36](#)). Data comparing hemodynamics on PAV with other modes are very limited. Patrick et al⁹² found that cardiac output was 22% higher during PAV than during VCV in eight patients with septic shock (see [Fig. 14-9](#) in ref. [93](#)). Blood pressure also was higher. These results likely were secondary to lower airway pressure, and hence mean intrathoracic pressure, during PAV promoting higher venous return. Kondili et al⁹⁴ compared hemodynamics on PAV and PSV when mean airway pressure was matched in the two modes. Cardiac index was slightly but significantly higher during PAV. There is no information on the effect of PAV versus other modes in patients with left-ventricular dysfunction. Such information is needed because, in theory, a lower intrathoracic pressure may not be helpful in these patients (higher afterload).

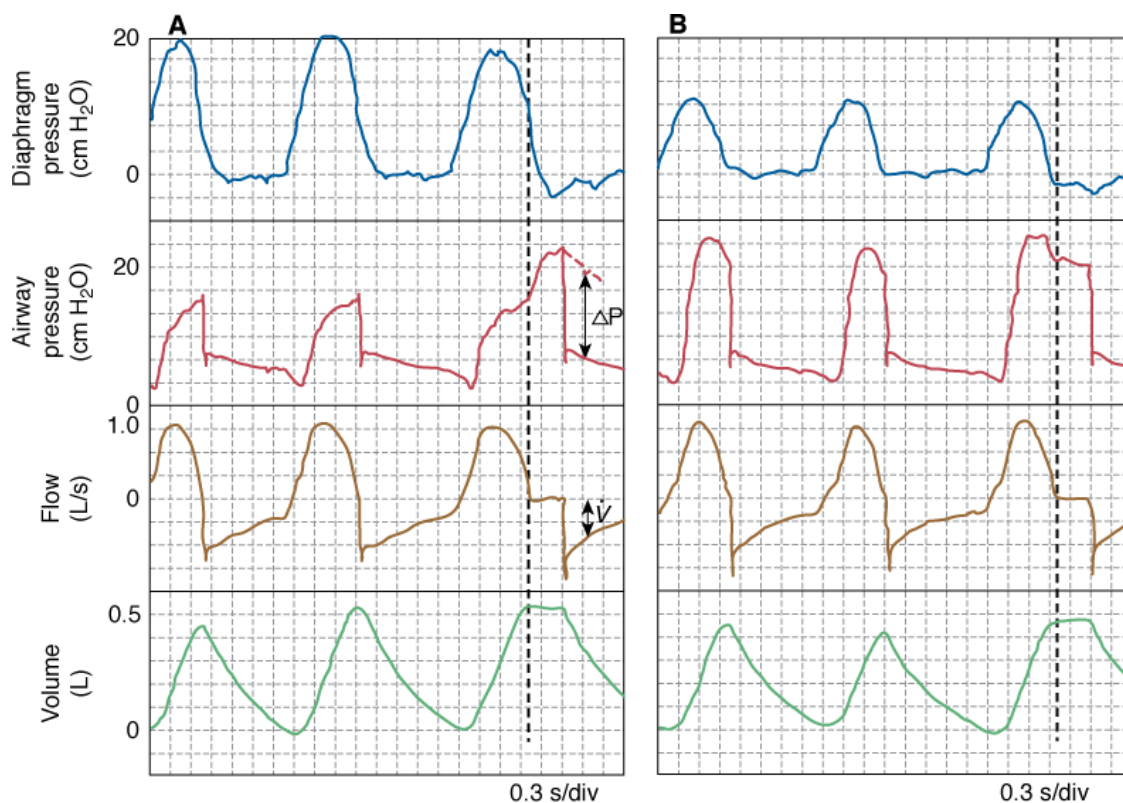
Clinical Consequences of the Physiologic Differences

Patient Management Issues

Noninvasive Monitoring of Respiratory Mechanics, Pmus, and Work of Breathing

PAV has unique features that allow estimation of passive mechanics noninvasively.^{56,57} In PAV, end of ventilator cycle occurs during the declining phase of inspiratory Pmus (see [Figs. 12-1](#) and [12-14A](#)). A brief end-inspiratory occlusion coincides with the terminal part of the declining phase and the decline of Pmus after the occlusion presents as a rise in airway pressure during the first part of occlusion ([Fig. 12-14A](#)). This phase ends shortly after the onset of occlusion, and its end is readily recognizable as the rise in airway pressure stops ([Fig. 12-14A](#)).⁵⁷ Paw at the end of this phase provides the passive recoil pressure associated with occluded volume. This is not the case with VCV or PSV because the end of the ventilator cycle may occur during an active inspiratory or expiratory phase. This approach has been validated,⁵⁷ and an automated version was included in new 840 PAV+ software (Covidien). A multicenter study confirmed its reliability ($r^2 = 0.92$ compared with elastance measured concurrently by esophageal catheter⁹⁵). A more recent study found a significant correlation between elastance measured on controlled mechanical ventilation and that measured during PAV.⁹⁶ There was, however, much scatter. This can be explained by the fact that, unlike the previous study,⁹⁵ the two measurements were separated by hours and were obtained with different modes and breathing patterns.

Figure 12-14

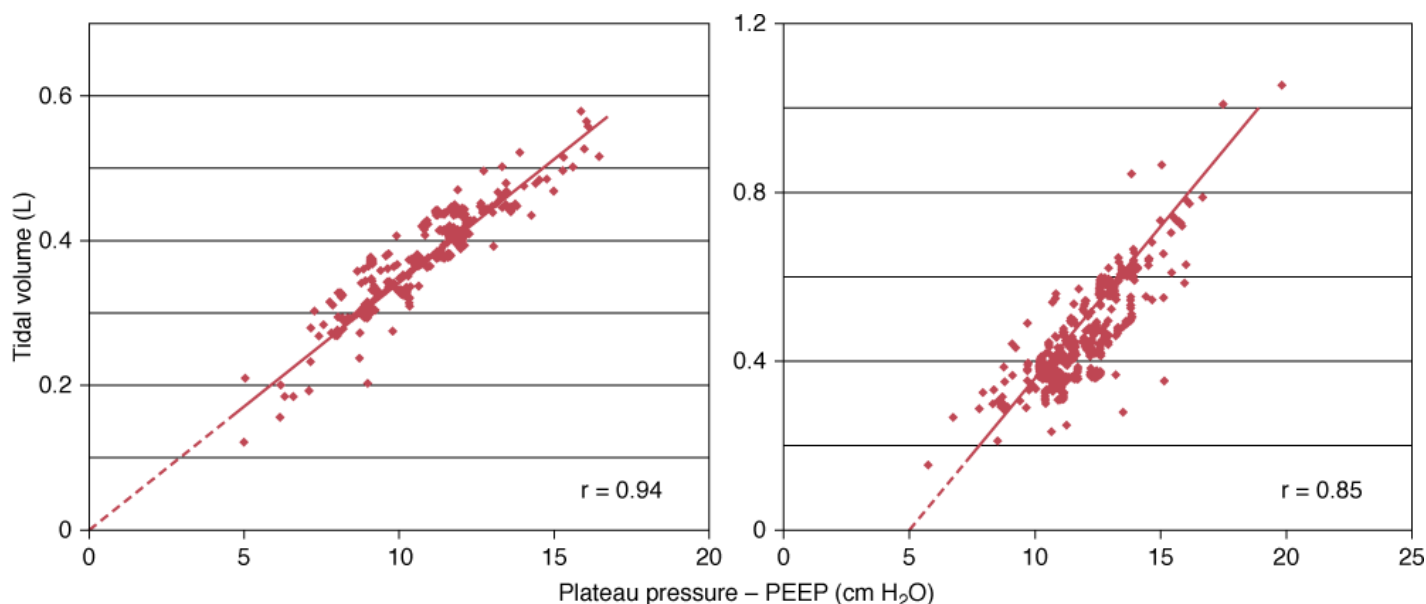


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Tracings from a patient on the 840 PAV+ option (Covidien) illustrating the random brief end-inspiratory occlusions. **A.** Forty percent assist. **B.** Eighty percent assist. Note the minimal change in breathing pattern and lower amplitude of diaphragmatic pressure swings. Airway pressure rises early during the occlusion in panel A, reflecting the fact the inspiratory phase ended during the declining phase of inspiratory pressure (*vertical dashed line*). During occlusion, airway pressure is inversely related to muscle pressure. By the end of occlusion, airway pressure had plateaued, corresponding to inspiratory pressure reaching its baseline level. In panel B, there is no rising phase during the occlusion because inspiratory pressure had returned to baseline already; at high assist, the inspiratory phase tends to end at a later point on the declining phase.⁷⁶ Note that airway pressure is approximately the same at end occlusion in both cases. Elastance is determined from $([\text{end-occlusion pressure} - \text{positive end-expiratory pressure}]/V_T)$. Expiratory resistance is determined from flow early in expiration and ΔP (the difference between airway pressure and elastic recoil at the same point). Elastic recoil is obtained from end-occlusion pressure minus an amount corresponding to the volume expired and elastance (*dotted line*).

Because V_T is usually variable on PAV, random end-inspiratory occlusions frequently produce a wide range of V_T s and plateau pressures from which confident estimates of the slope and pressure intercept can be obtained (Fig. 12-15). A positive intercept indicates either dynamic hyperinflation or a respiratory system that is stiffer in the lower part of tidal ventilation (see “Nonlinearity in the Pressure-Volume Relationship within the Tidal Volume Range” below). Because both abnormalities respond to increasing positive end-expiratory pressure (PEEP), changes in intercept can be used to set the PEEP level associated with best elastance. This process (display of plateau pressure vs. V_T) can obviously be incorporated in PAV delivery systems to help with the management of dynamic hyperinflation.

Figure 12-15



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Relationship between occlusion pressure and tidal volume. Data collected over approximately an hour of recording with randomly applied occlusions. The range of volumes was the result of spontaneous tidal volume variability. In both cases, a highly significant correlation was obtained. **A.** Pressure intercept not different from positive end-expiratory pressure (PEEP), indicating no dynamic hyperinflation. **B.** Patient with dynamic hyperinflation. Note the positive intercept.

Because Paw at the end of brief occlusions reflects passive recoil at end inspiration, it is possible to estimate passive *expiratory* resistance⁹⁷ (see Fig. 12-14A). This approach was incorporated in the 840 PAV+ option (Covidien). Its main limitation is that expiratory resistance in the PAV+ option is measured around the time of peak expiratory flow. Although this is acceptable in patients without severe COPD, it results in *underestimation* of resistance in COPD patients who display a sharp expiratory flow spike in early expiration (see, e.g., Fig. 12-5A). This early flow spike incorporates a large component derived from central airway collapse at the beginning of expiration and does not reflect the flow of air from the alveoli.^{98,99} Because alveolar flow is overestimated at this point, and flow is the denominator of the resistance equation, resistance can be markedly underestimated in the presence of this artifact. On the other hand, if the early expiratory flow spike is avoided, and resistance is measured later in expiration, resistance may be overestimated because of flow limitation. Thus, in the presence of severe expiratory flow limitation with an early flow spike, adjustment of PAV assist using an expiratory resistance value is not recommended. An alternate method that estimates inspiratory resistance from brief pressure pulses during inspiration has been described and validated.⁵⁶

Apart from overcoming the major practical limitation to implementing PAV, namely, knowing passive mechanics, continuous monitoring of passive R and E should help in monitoring disease progression and in the timely identification of complications (e.g., changes in lung water, accumulation of secretions, and bronchospasm). If a patient's condition deteriorates, it should be possible to sort out what happened to mechanics by observing recent trends in R and E. Furthermore, when R and E are known, it is possible to calculate patient-generated pressure (Pmus) and work of breathing in real time. Real-time estimates of Pmus can be used to trigger the ventilator, thereby eliminating the major remaining technical problem with PAV, namely delayed triggering in the presence of severe hyperinflation.

Improved Reliability of Ventilator Rate as a Measure of Distress and Weaning Failure

As indicated earlier (under “[Physiologic Consequences of Operational Differences](#)”), an increase in ventilator rate is not specific to distress in PSV and VCV. In either mode, a simple increase in effort, occurring spontaneously or as result of reduction in assist, may decrease ineffective efforts and result in an artifactual, sometimes dramatic increase in ventilator rate (see Fig. 12-8). Although this artifact can be identified from flow and Paw tracings,³⁴ considerable expertise is required. Furthermore, even if true RR is counted and found to have increased, the increase may be reflexive and not secondary to true distress (see “[Respiratory Rate in Different Modes](#)” above). With PAV, ineffective efforts are very rare, and inflation extends minimally into neural expiration. Accordingly, ventilator rate faithfully reflects patient rate.^{20,34,96} When it increases, distress can be inferred more reliably. Because a substantial increase in ventilator rate following a reduction in assist is used commonly to infer continued ventilator dependence, this feature of PAV should reduce false weaning failure verdicts.

The occurrence of an absolute RR of more than 35 breaths/min during a weaning trial is used commonly as a sign of weaning failure.^{102,103} Yet some patients breathe at rates greater than 35 breaths/min, even when assist is very high and efforts are very weak (see Fig. 12-8A).^{34,100} Thus, in some patients, a high absolute RR need not indicate distress. PAV can be used to determine whether absolute tachypnea observed during a weaning trial is distress-

related. Failure of RR to decrease as PAV assist is increased to high levels would indicate that tachypnea is not distress-related (see “[Respiratory Rate and Breathing Pattern](#)” above). Such a test would not be feasible with other modes because in such patients (high intrinsic RR), ventilator or even patient rate invariably will decrease as assist increases because of nonsynchrony. It remains to be determined, however, whether tachypnea that is not relieved by ventilator support is a reason to continue mechanical ventilation.

Choice of Ventilator Settings

With VCV and PSV, one does not know what is appropriate for each patient. Given what we know now about variability in undistressed (preferred) breathing pattern and time-to-time changes in demand, no general guidelines will be suitable for all patients; in most cases, the assist delivered will exceed what is necessary (see “[Tidal Volume in Different Modes](#)” above). With PAV, there is no uncertainty about what the patient wants or needs or how to set the level of assist. There is only one variable to consider (percent assist), and when percent assist is increased to a point where V_T and RR no longer respond to further increases, the values observed are what the patient needs.

This apparent simplicity of setting PAV was mitigated early on by a number of practical limitations, which proved problematic when simple PAV delivery systems were used (see “[Limitations](#)” below). These included the need to measure respiratory mechanics frequently and frequent alarming when the patient takes a large tidal volume or during runaways. To compound matters, troubleshooting these problems requires considerable expertise. These problems have now been largely overcome through introduction of methods for continuous monitoring of respiratory mechanics during PAV application, modification of alarm systems to allow for the spontaneous variability of tidal volume during PAV, and measures to avoid and limit runaways. A recent study¹⁰¹ on critically ill patients compared the number of interventions (ventilator settings and sedatives, analgesics and vasoactive medication dose manipulations) when patients were on PSV with the number while using a PAV system equipped with the above features (PAV+, Covidien). The number of interventions was significantly less on PAV+.

Clinical Outcome

It is reasonable to inquire as to whether the physiologic advantages of PAV translate into better clinical outcome. Unfortunately, information about outcomes is very scanty. This is so chiefly because, until very recently, available ICU ventilators (Winnipeg ventilator and Dräger Evita) were not equipped with means to provide smooth, nuisance-free PAV delivery over the extended periods required for outcome studies. It is hoped that with newly available systems such studies will be carried out. In the meantime, one is limited to speculation about how physiologic advantages may improve outcome.

Noninvasive Ventilation

Two studies compared outcomes with PSV and PAV in acute respiratory failure.^{38,39} Both found greater comfort and acceptance rate and a lower incidence of facial ulcers and conjunctivitis with PAV. One study found faster improvement in respiratory rate and P_{aCO_2} with PAV.³⁸ Neither study found a difference in intubation rate. Both studies, however, were seriously underpowered in this respect. When faced with a choice between severe distress and a somewhat uncomfortable way to relieve it, most people will opt for relief of dyspnea. Thus, nonacceptance rate with PSV is low (15% vs. 3% in PAV³⁹). Furthermore, intubation rate in such patients is also low (approximately 20% to 30%^{38,39}). It is estimated that more than 1000 patients are required to determine whether a new intervention reduces rate of intubation.³⁹ Rusterholtz et al randomized thirty-eight patients with acute cardiogenic pulmonary edema to receive noninvasive PAV or continuous positive airway pressure.⁴¹ They found no difference in failure rate (seven in each group) as defined by the onset of predefined intubation criteria, severe arrhythmias, or patient refusal. Again, the study size is insufficient to establish differences in clinical outcome.

Clinical Outcome in the Intensive Care Unit Setting

PAV may improve clinical outcome in a number of ways:

1. *Sedative use.* Most ICU patients are heavily sedated if they are not spontaneously unconscious. Sedatives may affect clinical outcome adversely.^{104,105} It is not clear to what extent patient-ventilator interactions contribute to need for sedation. To the extent that they may, PAV may result in less sedative use and reduction in sedation-related complications.
2. *Impact on sleep.* Poor patient-ventilator interaction may affect sleep adversely in the ICU.^{58,106} Sleep deprivation may increase blood pressure, depress immune function, and promote a negative nitrogen balance, actions that can affect morbidity adversely (for review, see ref. 106). By improving patient-ventilator interaction, PAV may help to reduce these complications. Bosma et al¹⁰⁷ monitored sleep for two consecutive nights in thirteen patients in the weaning phase. PAV and PSV were each used on one night and the order was randomized. The number of arousals and awakenings per hour of sleep was significantly lower on PAV and there was significantly more slow wave (deep) and rapid eye movement sleep. Interestingly, the number of arousals correlated with number of asynchrony events, which was greater with PSV.
3. *Barotrauma and ventilator-induced lung injury.* Excessive lung distension may result in further lung injury (see [Chapter 44](#)) and possibly multisystem organ failure¹⁰⁸ (see [Chapter 42](#)). There are a number of reasons why V_T will, on average, be smaller in PAV (see “[Tidal Volume in Different Modes](#)”

above). Furthermore, neural reflexes terminate inspiratory muscle activity if lung distension exceeds a certain threshold that is well below physiologic total lung capacity (TLC) (e.g., Hering-Breuer reflex). For example, when tidal expansion approaches TLC during exercise, further increases in ventilation are achieved automatically via increases in RR.¹⁰⁹ Because the assist terminates automatically with PAV when respiratory muscles are inhibited, overdistension beyond physiologic TLC (transpulmonary pressure approximately equal to 40 cm H₂O) is virtually impossible (except in patients with denervated lungs, such as after bilateral transplantation). Figure 12-16 shows average plateau pressure on PAV in forty-eight patients with a wide range of elastance. Plateau pressure was less than 30 cm H₂O above PEEP in all, and less than 20 cm H₂O above PEEP in all but four. These findings were confirmed in another recent study.⁹⁶ Considering that plateau pressure also includes chest wall recoil, lung distension was even less. It is reasonable to expect that this behavior will reduce barotrauma. It also may make it possible to achieve the objectives of permissive hypercapnia (small V_T) without the need for sedation or even hypercapnia (because RR increases automatically as end-inspiratory volume approaches TLC).

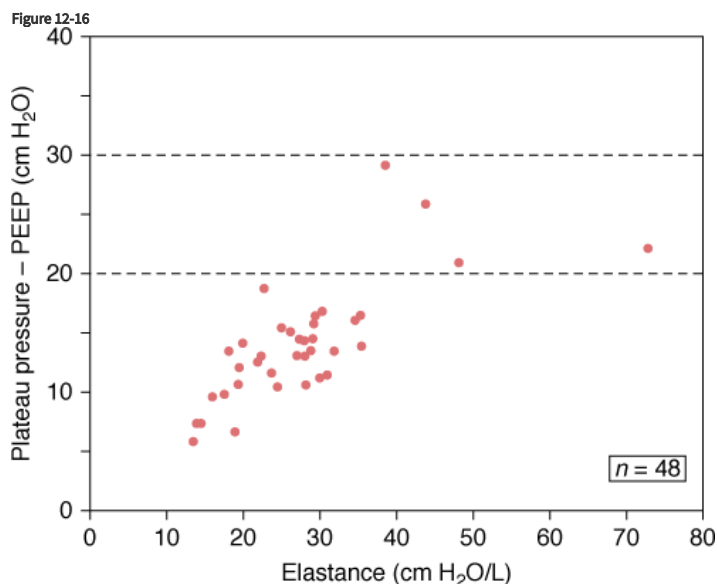
4. **Weaning.** The greater reliability of ventilator rate as a measure of distress (see “[Improved Reliability of Ventilator Rate as a Measure of Distress and Weaning Failure](#)” above) should decrease instances of false ventilator dependence. Tachypnea as the sole reason for a declaration of weaning failure accounted for 37% of all weaning failures in two large trials^{102,103} (A. Esteban, personal communication). It is tempting to speculate that some of these patients were not ventilator-dependent, so their identification under PAV will reduce ventilator time.

With PSV and VCV, PaCO₂ may decrease to very near the AT, resulting in extremely weak efforts, particularly during periods of reduced demand (see “[Physiologic Consequences of Operational Differences](#)” above). With VCV in the assist-control mode, it is also possible to produce protracted apnea. With PAV, a modest to moderate level of activation is present at all times, including during sleep. The likelihood of disuse atrophy of respiratory muscles (see [Chapter 43](#)) may be less, and this may facilitate weaning.

Most ineffective efforts occur during mechanical expiration. Accordingly, the inspiratory muscles are being lengthened during their activation. This type of contraction has been associated with muscle injury.^{110,111} Because ineffective efforts are very common with PSV and VCV⁸⁴ but not with PAV, this type of injury may be mitigated.

5. Continuous noninvasive monitoring of passive mechanics, made possible by PAV, may lead to better PEEP management and early detection and management of complications.

To date there has been only one study that compared clinical outcome with PSV and PAV in critically ill ventilated patients.⁹⁶ Xirouchaki et al⁹⁶ randomized 208 critically ill patients, mechanically ventilated on controlled modes for at least 36 hours, to receive either PSV (n = 100) or PAV + (n = 108). Specific written algorithms were used to adjust the ventilator settings in each mode. The patients were observed for 48 hours following transition from controlled ventilation unless they met predefined criteria either for switching back to controlled modes (failure criteria) or for breathing without ventilator assistance. Failure rate was significantly lower with PAV+ (11.1% vs. 22.0%, P = 0.040, odds ratio [OR] 0.443, 95% confidence interval [CI] 0.206 to 0.952) and the proportion of patients exhibiting major dyssynchronies was also significantly lower (5.6% vs. 29.0%, P < 0.001).



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Average plateau pressure (minus PEEP) in forty-eight patients with a wide range of elastance.

Commercially Available Proportional-Assist Ventilation Delivery Systems

The BiPAP Vision (Respironics) is the only ventilator currently suitable for noninvasive use. The Vision is equipped with leak-compensation algorithms. It also has excellent response characteristics and trigger sensitivity and has performed very well in several clinical trials.^{21,23,38,39} The Vision offers pathology-specific default startup settings as well as custom settings.

Until recently, the Evita ventilator (Dräger) was the only commercial ICU ventilator capable of delivering PAV. It is a basic PAV system with no capability to monitor passive mechanics in real time and with standard alarms. As indicated earlier, such basic systems (including the Winnipeg ventilator) have proven difficult to use except by experts and for short periods.

Covidien's 840 ventilator currently has a PAV+ option. The ventilator monitors mechanics continuously by applying random brief end-inspiratory occlusions (see Fig. 12-14). The results have been validated.⁹⁵ In addition, algorithms were added to limit the maximum elastic assist to a set level and the alarms were modified to allow for pressure and volume limits to be reached occasionally without the alarms sounding off. With this option, the maximum possible airway pressure, including PEEP, is 35 cm H₂O. This may be a limitation in a few patients receiving high levels of PEEP whose resistance is also high. Another limitation is that the resistance used to adjust the flow assist during inspiration is measured around peak expiratory flow. In the presence of an early expiratory flow spike (as in very severe COPD), this measurement can greatly underestimate the patient's true resistance and result in underassist (see "Noninvasive Monitoring of Respiratory Mechanics, Pmus, and Work of Breathing" above).

Limitations

The following mechanisms may result in excessive sounding of alarms and occasional instances in which patients are in distress despite a high percent assist. Fortunately, the underlying reasons for these difficulties are well understood, and auxiliary algorithms have become available that should mitigate most, if not all, of these problems.

Runaway Phenomenon

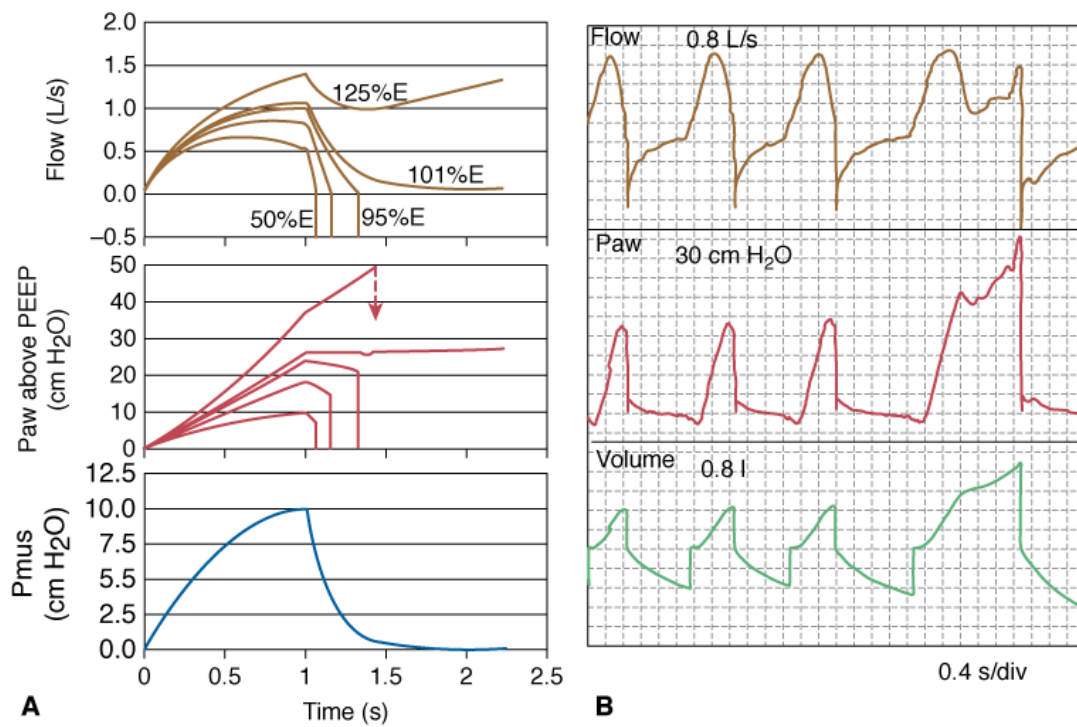
Despite its ominous-sounding name, the runaway presents no danger in modern ventilators with properly set pressure and/or volume limits. It is simply a nuisance in that, when it happens, it can trigger alarms, which can be annoying to staff and may promote anxiety in alert patients. Nonetheless, it is important to understand its mechanism and recognize its features, as this will be useful in managing PAV in patients who require a very high level of assist. The nuisance factor is much reduced in delivery systems with smart alarms, which allow for pressure and volume limits to be reached occasionally without the alarms sounding off.

Runaway occurs when the volume assist component ($\% \text{assist} [\text{estimated } V \times \text{estimated } E]$) exceeds the actual elastic recoil pressure of the respiratory system ($\text{actual } V \times \text{actual } E$) and/or the flow assist component of the assist ($\text{percent assist} [\text{flow} \times \text{estimated } R]$) exceeds the actual pressure dissipated against respiratory resistance ($\text{actual flow} \times \text{actual } R$). In modern PAV delivery systems in which the percent assist is used to adjust PAV, the percent assist cannot exceed 100. Thus, runaway occurs when flow and its integral, volume, are overestimated (e.g., uncompensated leaks) and/or estimated E and/or estimated R are greater than the actual values.

The runaway pattern depends on whether the resistive or elastic component is overassisted. With elastic overassist, the elastic pressure provided exceeds actual elastic pressure by an amount that increases as a function of volume. If flow is not overassisted, the resistive pressure provided is less than actual resistive pressure by an amount that is related to flow. So long as excess elastic pressure is less than the deficit in resistive pressure, a runaway does not occur because total applied pressure (P_{aw}) is less than the sum of actual resistive and elastic pressures. Runaway occurs when excess elastic pressure cannot be absorbed by the deficit (or reserve) in resistive pressure.

With elastic overassist, excess elastic pressure increases progressively during inspiration because volume rises throughout. By contrast, reflecting flow pattern, deficit/reserve in resistive pressure is highest in early and middle inspiration and decreases later. The point at which elastic overassist will exceed the reserve in resistive pressure (i.e., runaway) necessarily will occur late in inspiration. When VA is just greater than E, the runaway will occur at the very end of the inspiratory phase when flow is near zero (Fig. 12-17A). This fact has been put to use to measure actual patient E in sleeping or obtunded patients.^{6,17,93} VA is dialed up in small steps until the characteristic runaway pattern appears (Fig. 12-17B). At this point, VA is just greater than E. The more the elastic overassist, the sooner, in the inspiratory phase, runaway will develop (Fig. 12-17). Likewise, runaway occurs earlier if the difference between actual R and percent assist $\times$ estimated R is small because the deficit in resistive pressure will be less and more readily overcome by excess elastic pressure.

Figure 12-17

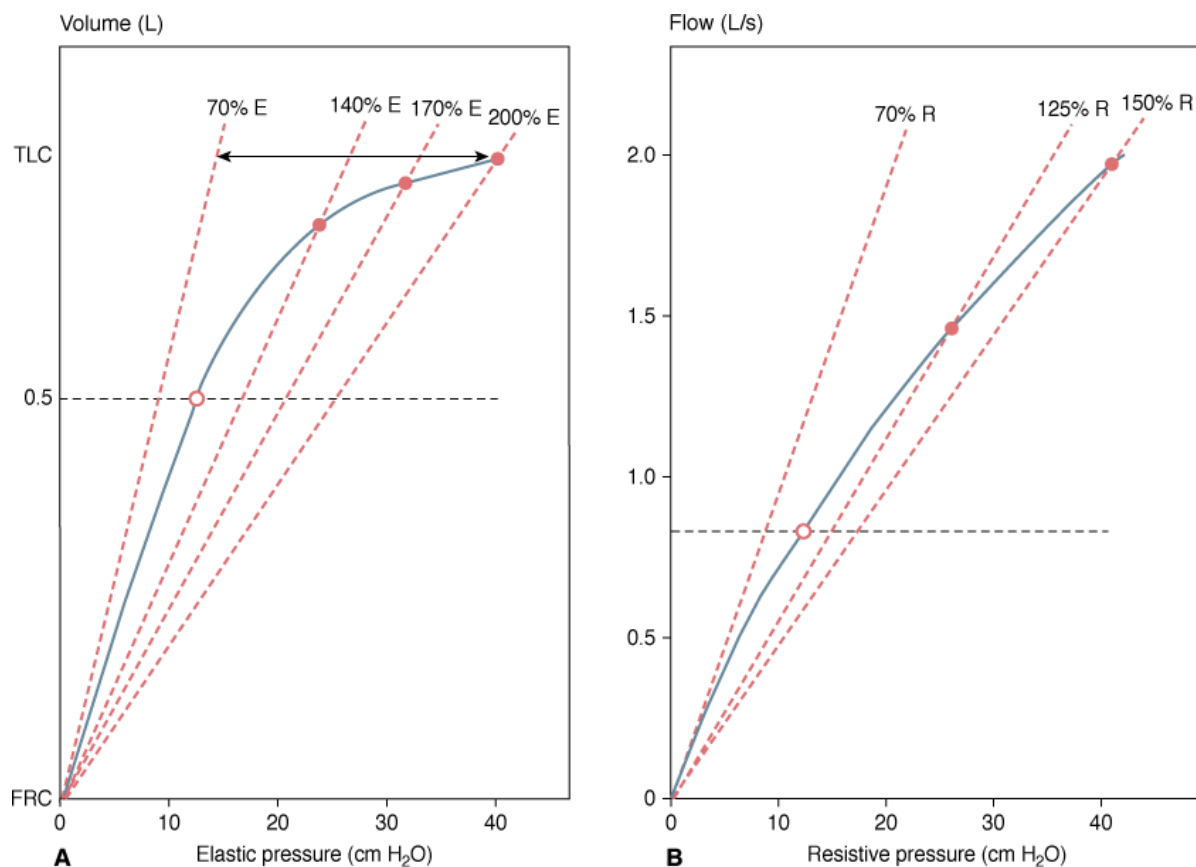


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Runaway. A. Model simulation of effect of increasing percent of elastance used for volume assist (VA). Note that as soon as VA exceeds elastance (101% E), flow fails to return to zero at the end of inspiratory phase. At higher levels, the runaway begins earlier, and flow and pressure increase progressively until the cycle is terminated by a physiologic or ventilator limit. **B.** Patient with COPD and moderate dynamic hyperinflation, 90% assist. Note the spontaneous occurrence of a runaway breath (last breath). The pattern is intermediate between the 101% and 125% in the *left panel*. Note that in the breath preceding the runaway, exhaled volume exceeded inhaled volume, suggesting less dynamic hyperinflation at the beginning of the runaway breath. See text for more details. The runaway breath was terminated by the ventilator's high-pressure limit.

Once a runaway develops as a result of elastic overassist pressure and volume will continue to rise beyond the end of inspiratory effort (see Fig. 12-17B) until the cycle is stopped by (a) a set pressure or volume limit, or (b) volume approaching the stiffer upper part of the pressure-volume curve of the respiratory system (Fig. 12-18A). The latter results in a progressive increase in actual elastance that cancels the overassist. Thus, the nonlinearity of the pressure-volume relationship of the respiratory system acts as natural protection against excessive overdistension.

Figure 12-18



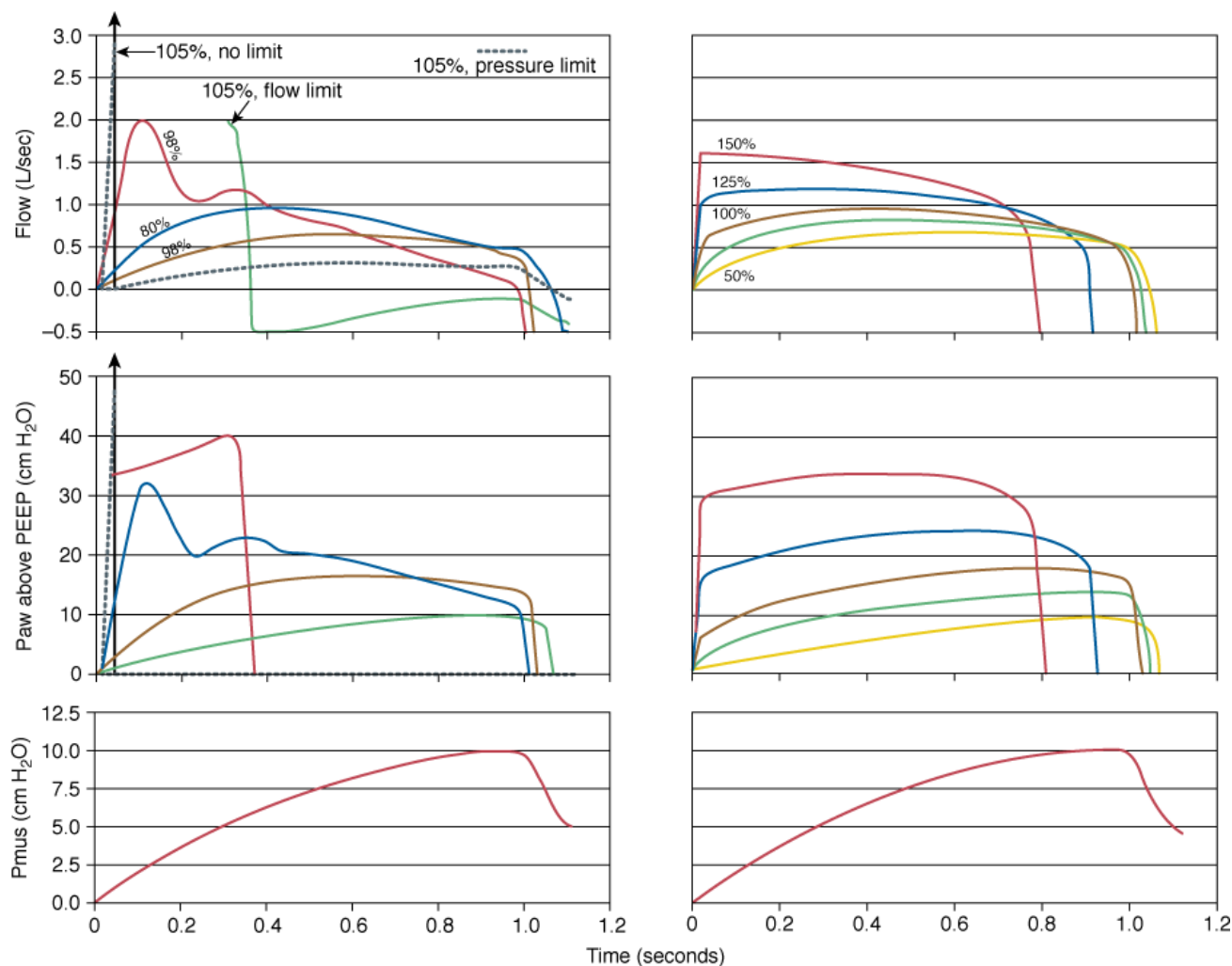
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Physiologic limits to runaway. **A.** Typical pressure-volume curve in an average ventilated patient. Elastance (E) at a tidal volume (V_T) of 0.5 L (the average V_T on PAV in this case) is 25 cm H₂O/L (open circle). Left diagonal line: Volume assist = 70% E . The elastic pressure delivered by the ventilator is less than elastic recoil at all volumes (no runaway). The next three diagonal lines represent progressively increasing overassist. An excess elastic pressure is delivered in the spontaneous tidal volume range (horizontal distance between diagonal lines and pressure-volume line), forcing a volume runaway. Because of the stiffening of the system near TLC, however, excess pressure decreases as volume increases. Runaway ends when the diagonal line meets the pressure-volume line because there is no longer an excess pressure. Note that even with a marked overassist (200% E), runaway stops at an elastic pressure of 40 cm H₂O, corresponding to a physiologic TLC value. **B.** Typical pressure-flow relationship in an average patient (airway resistance is 10 cm H₂O/L/s with a no. 8 endotracheal tube). Total resistance at a flow of 0.8 L/s is 15 cm H₂O/L/s. A flow assist in excess of this value will cause a flow runaway (right two diagonal lines). Because the pressure-flow relationship is not linear, excess resistive pressure decreases as flow increases. Flow runaway stops when the diagonal line meets the pressure-flow line. Excessive overdistension of the lung is also mitigated in this case by nonlinearity of the pressure-volume curve. Note that if VA is less than E (left diagonal line, panel A), the deficit in elastic pressure increases dramatically near TLC (horizontal arrow, panel A). This helps offset the excess resistive pressure and aborts the flow runaway below physiologic TLC. *FRC*, functional residual capacity; *TLC*, total lung capacity.

By analogy, with resistive overassist, runaway will occur when the excess resistive pressure provided by the ventilator exceeds the deficit/reserve in applied elastic pressure. This point invariably will occur at the very beginning of inspiration, where volume, and hence elastic assist, is near zero, and there is no possibility for the elastic deficit/reserve to absorb the excess resistive pressure.

The pattern of flow runaway depends on whether the pressure-flow relation is linear (Fig. 12-19). A linear pressure-flow relation occurs when breathing via the mouth, thereby excluding the nonlinear nasal pressure-flow relation, or when the nonlinear relation of the endotracheal tube is offset independently by automatic tube compensation (ATC). Here, as flow assist just exceeds actual resistive pressure, there is a very rapid increase in flow and pressure that can be aborted only by a ventilator limit (see Fig. 12-19A). The inflation phase is very brief. By contrast, when the pressure-flow relation is nonlinear, there is no discrete change as FA exceeds R . Rather, change is gradual and consists of a progressive shift in peak flow to earlier points in inspiration (see Fig. 12-19B). Flow pattern at high levels of overassist resembles that of PSV. The rapid increase in flow early in inspiration may truncate the breath, however, and sound an alarm if peak pressure limit is reached.

Figure 12-19



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Patterns of flow runaway. Percent denotes ratio of flow assist to resistance at a reference flow (0.5 L/s in panel B). **A.** Linear pressure-flow relationship. FA is increased in steps beginning with 50% R. As the assist approaches 100% R, some oscillations become apparent. At just above 100% R, there is a sudden change in response. Without any ventilator limits, flow and pressure increase very rapidly to extremely high levels at the beginning of inspiration. Termination of runaway depends on ventilator limits. With activation of peak-pressure limit, the cycle is aborted rapidly. If the maximum flow capability of the ventilator (e.g., 2 L/s) is reached before the peak-pressure limit, the cycle continues a while more at the maximum flow and then self-terminates rapidly. With a constant resistance, therefore, transition of assist to above 100% is abrupt. **B.** Nonlinear pressure-flow relationship (e.g., in the presence of an endotracheal tube). Runaway is blunted by the increasing resistance at higher flow (see Fig. 12-18B). There is no clear change in response at 100% assist. Rather, change is gradual and consists of a progressive shift in peak flow to earlier points in inspiration. *Paw*, airway pressure; *Pmus*, respiratory muscle pressure.

The development of algorithms to continuously monitor mechanics and their incorporation in commercial PAV delivery systems has greatly reduced this problem when such systems are used.

Runaways need not occur on every breath (see, e.g., Fig. 12-17). Because true elastance (E) may vary breath by breath (see “[Dynamic Hyperinflation](#)” and “[Nonlinearity in the Pressure-Volume Relationship within the Tidal Volume Range](#)” above), estimated E may exceed actual E in some breaths, even though it is less than average E . Assume that average E , determined from a number of end-inspiratory occlusions, is 30 cm H₂O/L, with a range of 24 to 36 cm H₂O/L. At 90% assist (i.e., 27 cm H₂O/L), some breaths may develop runaway. The higher the percent assist, the more frequent are the runaways.

Accuracy and Stability of Respiratory Mechanics Values

Knowledge of R and E is not necessary in noninvasive applications. Feedback from the alert patient ensures that VA and FA are appropriate. In the usually obtunded ICU patient, subjective feedback is not possible, and setting PAV properly requires knowledge of passive mechanics. This has created problems for two reasons: first, measuring passive mechanics in the usual way (under sedation, hyperventilation, and/or paralysis) is cumbersome, requires expertise, cannot be done frequently, and the results may not reflect the E and R values on PAV (because of differences in V_T , flow rate, dynamic hyperinflation, and so on). Second, passive mechanics frequently change (see “[Response to Changes in Respiratory Mechanics](#)” above). Values that are accurate at one point may become inaccurate later.

When differences between actual and assumed R and E values are small (e.g., <25%) and percent assist is not high, the impact is operationally insignificant. For example, with a 20% error at 50% assist, actual assist will be 40% or 60% depending on the direction of the error. Either value is consistent with proper functioning, and if the assist is insufficient, there is room to increase it. By contrast, with a 50% error at 80% assist, actual assist will be 40% or 120%. In the former case, assist may be insufficient, resulting in distress when the patient is “supposedly” receiving high assist, and there is little room for further increases. In the latter case, runaway will develop, with frequent sounding of the alarms, even though it should not happen (percent assist <100). In my opinion, the discrepancy between assumed and actual R and E is the most important source of implementation difficulties. Fortunately, this has now been largely resolved (see “[Noninvasive Monitoring of Respiratory Mechanics, Pmus, and Work of Breathing](#),” and “[Commercially Available Proportional-Assist Ventilation Delivery Systems](#)” above).

Leaks

Leaks affect PAV delivery in much the same way as overestimation of E and R does. For example, if half the gas leaving the ventilator leaks out, assist will be twice that intended, or analogous to 100% overestimation of E and R. Thus, the magnitude of overestimation is related directly to the magnitude of the leak. As in the case of overestimation of E and R (see preceding section), a small leak will not have much impact if percent assist is low or moderate and will cause runaway if assist is high. Large leaks would result in runaway at all but the lowest assist levels.

In noninvasive applications, leaks can be huge. For this reason, noninvasive PAV delivery systems must be equipped with leak-compensation algorithms. In the ICU setting, leaks are usually very small, except in bronchocutaneous fistulas. Nonetheless, checking for leaks should be undertaken if alarms begin sounding excessively when they did not before.

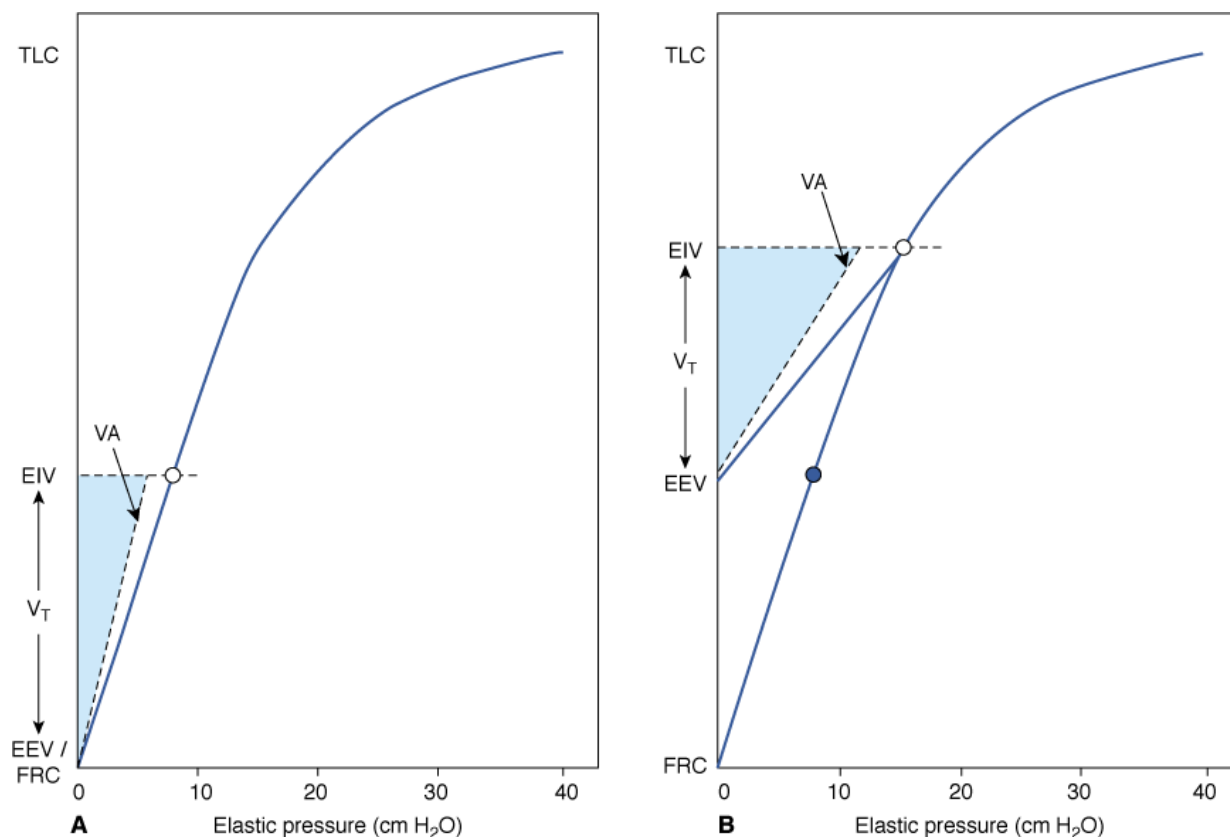
Inclusion of automatic mechanics in PAV delivery systems essentially should eliminate leak-related problems. Thus, when a leak exists, the end-inspiratory occlusion technique (see [Fig. 12-14](#)) will underestimate E by an amount corresponding to the leak (provided inspired volume is used to compute E). This should offset the error at the ventilator level. Likewise, the pulse technique (resistance)⁵⁶ will underestimate inspiratory resistance in the presence of leaks, thereby offsetting the pressure-delivery error at the ventilator level.

Dynamic Hyperinflation

Dynamic hyperinflation (DH) presents occasional implementation difficulties, particularly when severe and associated with marked respiratory muscle weakness. Thus:

1. By definition, DH means that elastic recoil pressure is greater than zero at inspiratory onset. Inspiratory muscles must generate enough pressure to offset this elastic recoil before the ventilator is triggered (see, e.g., [Figs. 12-7 and 12-8](#)). Therefore, by the time the ventilator is triggered, a finite fraction of the patient's inspiratory phase will have elapsed. This delay is a function of magnitude of DH and the rate of rise of Pmus (see [Fig. 12-7](#)). When DH is large (e.g., 10 cm H₂O in [Fig. 12-7A](#)) and the rate of rise of Pmus is low, triggering may not occur until near the end of inspiratory effort (see, e.g., [Fig. 12-7A](#)). In PSV and VCV, once triggering occurs, a substantial breath will be delivered. Although volume delivery will be almost entirely during the patient's expiratory phase, adequate ventilation nonetheless will be delivered, making it possible for chemical drive to remain low. In PAV, the ventilator will provide support only for the remaining duration of inspiratory effort. When triggering is much delayed, the duration of support may be very brief, resulting in inadequate ventilation (see, e.g., [Fig. 12-13](#)). The only way the patient can receive reasonable ventilation is to increase the rate of rise of inspiratory effort in order to advance triggering (see, e.g., [Fig. 12-7B](#)). This will occur naturally at the expense of a higher chemical drive. When respiratory muscle reserve is high, the required increase in respiratory muscle output may be tolerated. When DH is very high and muscle reserve is quite low, distress may develop.
2. As illustrated in [Figure 12-20](#), the effective assist received is less than the percent assist dialed in. This is so because the ventilator is unaware of the Pmus generated before triggering; the assist is applied only to that part of Pmus in excess of that required for triggering. Thus, not only does the patient have to generate more pressure to advance triggering, but the maximum percent assist that can be delivered effectively also is less. Distress may occur even at the highest possible assist.
3. Even if an assist level is found with which the patient is comfortable, runaway still may develop in occasional breaths, causing confusion (as percent assist is technically less than 100% E). The duration of expiration of spontaneous efforts can be quite variable. A breath preceded by a long expiratory time begins at a lower absolute volume (less DH). The same percent assist that was not associated with runaway when end-expiratory volume was higher may now cause runaway (see [Fig. 12-17B](#)). This can be appreciated by moving the VA line downward in [Figure 12-20B](#). It will be noted that at some point the VA line will cross to the right of the pressure-volume line (i.e., overassist).

Figure 12-20



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Impact of dynamic hyperinflation (DH) on PAV delivery. **A.** No DH (FRC and end-expiratory volume are the same). If 80% assist is dialed, the elastic work done by the ventilator (*shaded triangle*) is 80% of the total elastic work (*sum of shaded and open triangles*). **B.** DH: elastic recoil pressure at end-expiratory volume is 8 cm H₂O (*solid circle*). Measured elastance ($E = \text{plateau pressure}/V_T$) overestimates actual elastance because V_T , the denominator, underestimates the difference between end-inspiratory volume and FRC (compare the solid diagonal line with the slope of the main pressure-volume curve in the V_T range). At 80% of this inflated elastance (*VA line*), the ventilator is providing only 50% of elastic work (compare the shaded triangle with the pressure-volume area in the V_T range). There is little room to increase the assist. Forcing *VA* to exceed measured elastance will result in runaway. In such cases, therefore, the maximum assist that can be delivered is limited. *EEV*, end-expiratory volume; *FRC*, functional residual capacity; *TLC*, total lung capacity; *VA*, volume assist; V_T , tidal volume.

The preceding account should not suggest that patients with severe COPD or other causes of DH are difficult to support. As judged by numerous reports^{19–21,24–26,36,39,40} and my own experience, the vast majority can be supported very comfortably. Even in patients with severe COPD, elastic recoil at end expiration is usually approximately 3 to 5 cm H₂O,^{114–116} a level that is accommodated easily by most patients. Nonetheless, a very high expiratory resistance combined with a high ventilatory demand and weak muscles may make it difficult to achieve adequate support.

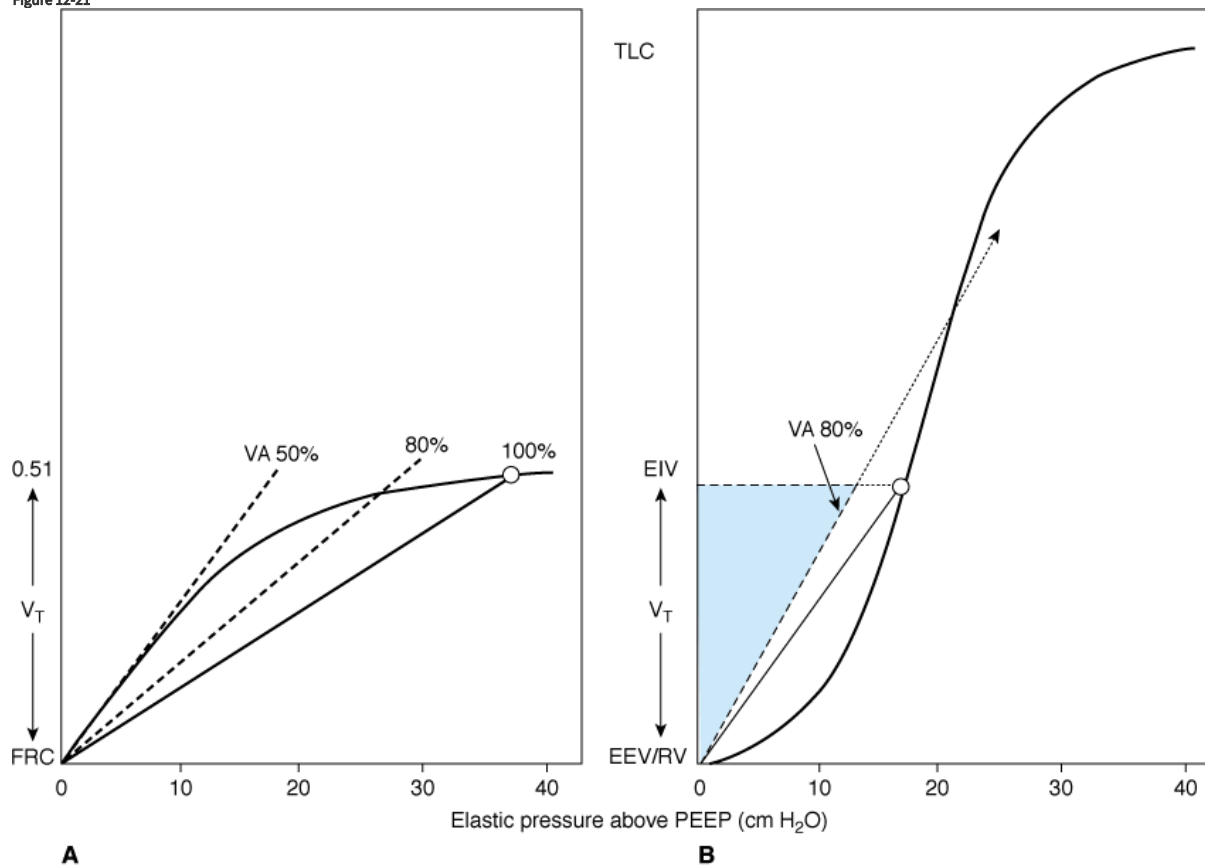
The adverse impact of severe DH on PAV's performance can be mitigated in some patients by increasing PEEP. This is effective, however, only when the high expiratory resistance is related to expiratory flow limitation in the lung. Where the high resistance is in the tubing, raising PEEP simply increases lung volume without relieving DH. The impact of DH can be eliminated completely if the ventilator is made to trigger at the onset of inspiratory effort instead of being triggered by inspiratory flow or airway pressure. Because it is possible to measure passive E and R during PAV, it is also possible to generate a continuous estimate of P_{mus} in real time. Such a signal can be used to begin the assist at the time of onset of inspiratory effort, instead of the onset of inspiratory flow. Another method for noninvasive identification of the onset of effort in real time has also been developed recently (see "effort channel" in Fig. 12-7).¹⁰⁰ With either system, PAV assist continues to be a function of flow and volume, but the reference flow is the expiratory flow at onset of effort (as opposed to zero flow, as is currently practiced). In this fashion, assist would apply throughout inspiratory effort.

Nonlinearity in the Pressure–Volume Relationship Within the Tidal Volume Range

The relation between volume and elastic recoil is sigmoid.¹¹² Most patients breathe in the linear midrange, and E is nearly constant within the V_T range. In others, the pressure–volume relationship is not linear within V_T . This creates some difficulties because there is no fixed E to use for the sake of setting the volume assist. The difficulties depend on whether V_T falls within the stiff upper range (high-end nonlinearity) or the stiff lower range (low-end nonlinearity).

1. **High-end nonlinearity** (Fig. 12-21A). This occurs when end-inspiratory volume approaches TLC, such as when external PEEP is excessive or in very severe restrictive disease (e.g., severe acute respiratory distress syndrome). Here, the maximum elastic assist that can be provided without runaway is limited, and the highest V_T , with or without runaway, is constrained by physiologic TLC (Fig. 12-21A). Should this V_T be inadequate for the patient's ventilatory demand, distress will develop and cannot be relieved by increasing percent assist. Unless heavily sedated, such patients (high demand plus very stiff respiratory system) require supraphysiologic distending pressures for adequate ventilation, and PAV is not suitable.
2. **Low-end nonlinearity** (see Fig. 12-21B). Here, the system is stiffest in the low range of tidal volume, for example, when end-expiratory volume is close to residual volume (abdominal distension and obesity¹¹⁸⁻¹²⁰) or when derecruitment of airways or alveoli occurs within the tidal volume range. The impact of this abnormality on PAV delivery is similar to that of dynamic hyperinflation (see "Dynamic Hyperinflation" above); the assist received by patient is less than intended (see Fig. 12-21B). When assist requirement is low, this behavior presents no difficulty. When assist level must be high (e.g., 80%), however, the patient may be comfortable, but because of the usually large breath-by-breath variability in V_T , runaways may occur during large breaths, triggering alarms (see Fig. 12-21B). Therefore, in some patients it may be difficult to reach an assist level that is both adequate and free of frequent alarming. This problem can be eliminated by increasing PEEP, thereby placing V_T in the linear pressure-volume range.

Figure 12-21



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Impact of nonlinearity in the pressure-volume relationship within tidal volume. **A.** High-end nonlinearity. The pressure-volume relationship becomes quite flat within 0.5 L of FRC. Elastance at a volume of 0.5 L is 75 cm H₂O/L (pressure and volume values at the open circle). The maximum VA that can be given without runaway is 50%. At higher assist, runaway develops until the flat region is reached. In either case, the maximum V_T that can be obtained with PAV is limited by physiologic TLC. **B.** Low-end nonlinearity. Elastance is determined at an average V_T given by the open circle. At a VA of 80% E, the elastic assist received is only 55% (compare the hatched area with the total area inside the pressure-volume line). Furthermore, if the patient makes a larger effort and obtains a larger V_T , elastic assist may exceed elastic recoil, and runaway may occur (arrow). At 50% assist, runaways would not occur, but the patient receives only 35% assist (not shown). *EEV*, end-expiratory volume; *FRC*, functional residual capacity; *TLC*, total lung capacity; *VA*, volume assist; *V_T*, tidal volume.

Ventilator Response Time

Response delays are unavoidable in electromechanical systems. Excessive response delay would result in less assist than intended as well as delays in cycling off. In the original Winnipeg ventilator, the delay was approximately 40 milliseconds. I had assumed that the delay simply would result in a parallel shift relative to the target Paw waveform⁹³ and thought this would be acceptable. However, Du et al⁷⁶ demonstrated that ventilator response delay

becomes compounded during the inspiratory phase, resulting in cycling-off delays that are longer than the nominal ventilator delay. In a recent study, cycling off delay while using the PAV+ option in ventilator-dependent patients averaged 0.25 second (range: 0.17 to 0.51 second).¹¹⁷ This is acceptable, particularly when considering that transition from inspiration to expiration does not normally occur at peak P_{mus} but during the declining phase of inspiratory activity,^{49,121–123} so that some delay is inevitable even without a ventilator. Nonetheless, that response delays become compounded during PAV delivery⁷⁶ emphasizes the need for fast-responding gas-delivery systems in this mode.

Excessive Alarming

Standard ventilator alarms that sound off every time a limit is reached proved to be a considerable nuisance in the PAV mode. Because of the spontaneous variability in breathing pattern, one or more limits may be reached frequently. Furthermore, without automatic mechanics, runaway breaths were frequent. With incorporation of automatic mechanics (see “[Noninvasive Monitoring of Respiratory Mechanics, \$P_{mus}\$, and Work of Breathing](#)” above), occurrence of runaway will be greatly reduced, but some may still occur (see “[Dynamic Hyperinflation](#)” and “[Nonlinearity in the Pressure–Volume Relationship within the Tidal Volume Range](#)” above), and spontaneous variability will remain as a possible source for activating alarms. Accordingly, alarms in the PAV mode should take into account the breath-by-breath variability in breathing pattern and mechanics. A reasonable approach (e.g., 840 PAV+ option) is to use special filters whereby ventilator limits still function as such for safety reasons, but alarms sound off only when the frequency of reaching a ventilator limit exceeds a threshold value.

Indications and Contraindications

PAV is appropriate to use in all but a few situations:

1. *Respiratory depression.* Safe use of PAV requires that the patient’s respiratory muscle output be responsive to changes in P_{aCO_2} , P_{aO_2} , and pH.

Absolute contraindications are patients with central apnea or very weak efforts and *no respiratory distress despite an abnormally low pH*. Patients who are intubated for respiratory depression (e.g., overdose or neuropathology) or who had to be heavily sedated for intubation should not be placed on PAV initially. The situation is different in patients already being ventilated with another mode. Here, weak efforts and even no efforts (e.g., on assist-control ventilation or having recurrent central apneas on PSV⁵⁸) are almost invariably secondary to overventilation. The assist with the other mode should be reduced first to establish that efforts resume at a reasonably normal pH before switching to PAV.

Care should be exercised when initiating PAV in a patient with known or suspected chronic CO_2 retention whose P_{aCO_2} and pH were normalized by another ventilation mode. Such patients often respond poorly to CO_2 until P_{aCO_2} reaches a level commensurate with their preintubation value. This may result in acute hypercapnia and acidemia following a switch to PAV. Chronic CO_2 retention per se is clearly not a contraindication, and many such patients were treated adequately with PAV. Patients who develop shallow, relatively slow breathing *with no distress* on PAV should have their blood gases checked, and if they are acidemic, they should be switched to another mode. These patients likely had undocumented prior chronic CO_2 retention.

2. *Need for heavy sedation:* The respiratory response to heavy sedation is unpredictable. When heavy sedation is required acutely for control of severe agitation or seizures, it is better to switch the patient to another mode until it is clear that the patient is making efforts at an acceptable pH.
3. *Severe neuromuscular weakness:* These patients may have difficulty triggering the ventilator and when triggering occurs it may be close to the end of inspiratory effort. Because with PAV, unlike other modes, the ventilator cycle terminates at the end of inspiratory effort, the inflation cycle may be very brief and ventilation may be inadequate.
4. *Bronchocutaneous fistulas.* Patients with bronchocutaneous fistulas should have their leak measured before switching to PAV. If the leak is high on another mode (e.g., exhaled volume less than 75% of inhaled volume), it is better not to use PAV unless the ventilator is equipped with automatic mechanics or leak-compensation algorithms.

Use of Sedatives in Patients on Proportional-Assist Ventilation

Sedatives are not contraindicated when PAV is used. When the response of a patient to sedatives is not known, the dose should be titrated initially until the amount required to attain the desired effect without severe depression is learned. Because backup systems always must be available on PAV delivery systems, the inadvertent use of an excessive amount of sedatives should not be hazardous.

Adjustment at the Bedside

Noninvasive Application

General instructions regarding noninvasive ventilation are similar to those for other modes and are not discussed here (see [Chapter 18](#)). The following procedure pertains to BiPAP Vision (Respironics).

Startup

It is neither necessary nor recommended to attempt to measure respiratory mechanics. Patients are alert, and settings are best accomplished using patient feedback. Before attaching the ventilator to the mask, the limits should be set initially to a maximum pressure of 20 cm H₂O, a maximum volume of 1.5 L, and a maximum inspiratory duration of 3 seconds. Expiratory positive airway pressure should be set to the minimum value (4 cm H₂O). Many patients prefer a lower expiratory positive airway pressure, but 4 cm H₂O is the lowest setting on the BiPAP Vision. Set the ventilator in the “custom” mode to permit manual adjustment. Percent assist should be set to 99%. In this fashion, the E and R inserted in the ventilator are the actual VA and FA delivered to the patient. Set E and R initially to 2 and 1, respectively.

After attaching the ventilator to the patient and waiting a few breaths for the patient to adapt, inform the patient that the assist will be increased in steps and that he or she should signal approval/disapproval by some agreed-on system. Increase E in steps of one, waiting a few breaths between steps, until the patient signals that the assist is too high. Go back one or two steps. Repeat using the R input. Increase expiratory positive airway pressure by 1 cm H₂O. If the patient signals approval, keep increasing expiratory positive airway pressure. Most will not want more. Finally, perform final adjustment of E. Some patients will prefer a somewhat lower E setting once the other variables are optimized.

Observe the screen display for the range of airway pressure and V_T. Set the high limits a reasonable amount above the observed ranges. For example, if peak Paw ranges up to 15 cm H₂O and V_T ranges up to 0.7 L, set the pressure limit to 20 cm H₂O and the volume limit to 1 L.

Subsequent Management

This is guided by clinical course. If distress progresses or reappears, the values of E and R should be retitrated. Nasal congestion should be avoided. Ask the patient if his or her nose is congested. If so, use decongestant, or switch to full face mask. If distress persists, then other methods of therapy should be considered. With clinical improvement, percent assist is reduced in steps guided by patient feedback. Because improvement may occur rapidly, it is a good idea to ask the patient at intervals whether the assist is too much and to see whether a lower percent assist is tolerated.

Troubleshooting

Although the ventilator corrects for leaks, it takes it three to five breaths to adjust to a leak change. A sudden change (e.g., mouth opening or loosening of mask) may result in a transient increase in applied pressure (up to the set limit). Patients should be instructed to keep their mouth closed. For talking or feeding, percent assist can be reduced to zero (or the ventilator disconnected) temporarily. When the problem occurs without mouth opening, the straps need to be tightened.

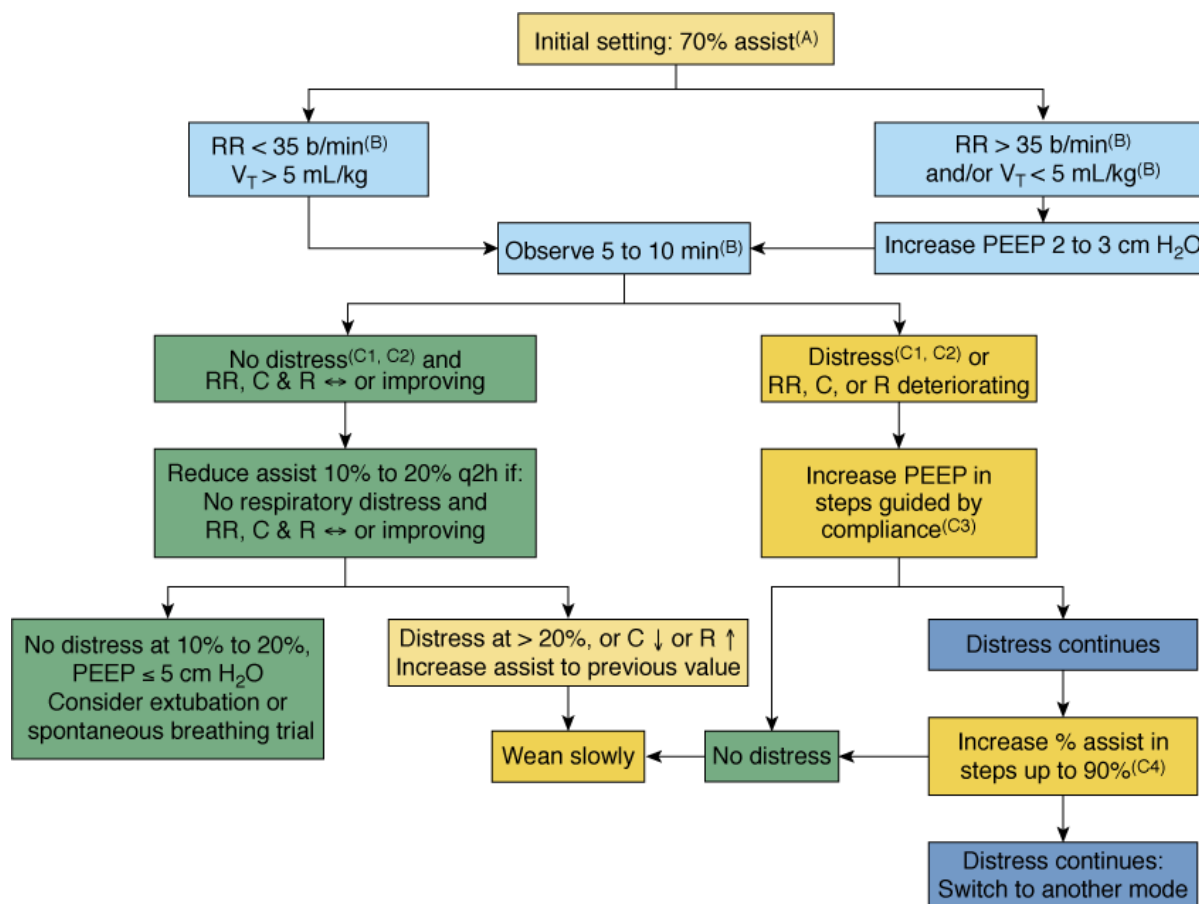
Frequent activation of alarms may be secondary to the alarm levels being set too low relative to the prevailing ventilatory demand ($\dot{V}_E$ and V_T demands may have increased) or to improvement of mechanics since the last titration, with the assist now becoming excessive. Note the range of V_T on the display, in particular the highest volumes. Reduce percent assist by 20% or so. If the large V_Ts are eliminated and alarming stops, the problem was an improvement in mechanics, and the large breaths were runaways. Retitrate E and R. Conversely, if the large volumes continue to occur, then the problem is large ventilatory demand or marked breath-by-breath variability. In such cases, the limits should be adjusted upward.

Intubated Patients

The following pertains to the PAV + option on the 840 ventilator because it is currently the only ventilator available with automatic mechanics and “smart” alarms. The procedure for setting basic PAV systems has been described elsewhere.¹²⁴ As indicated earlier, such systems are difficult to manage for extended periods. If other systems are used, *full* (i.e., 100%) automatic tube compensation should be avoided. With complete compensation for the nonlinear component of resistance, very small errors in estimated R may cause aggressive flow runaway with rapid loss of the assist secondary to ventilator limits (see “Runaway Phenomenon” above).

The following algorithm for start up and subsequent management (Fig. 12-22) was developed jointly by the author and Dr. D. Georgopoulos, based on experience with hundreds of patients with various pathologies. It was originally published in the *International Journal of Intensive Care*¹²⁵ and partly reproduced in *Essential Practices in Respiratory Care*.¹²⁶ The reader is alerted to errors in the flowcharts printed in these two publications which, accordingly, should not be followed.

Figure 12-22



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Proposed algorithm for PAV management. C, compliance; R, resistance; RR, respiratory rate; VT, tidal volume. Superscripts refer to paragraphs in the text (under “adjustments at the bedside”: Intubated Patients) with important clarifications.

Initial Ventilator Settings:

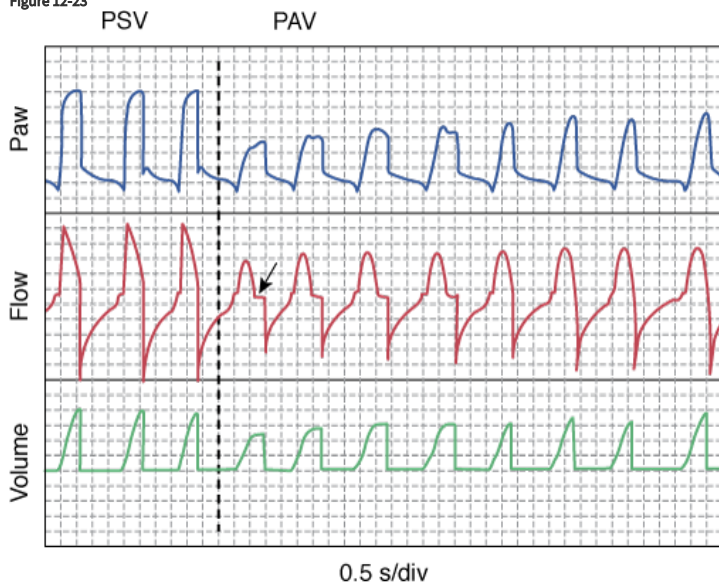
1. Enter ideal body weight (IBW). *This is an important step.* IBW does not have to be precise but should be a reasonable approximation of reality. It takes the ventilator four breaths to determine actual E and R. In the interim, it uses default values based on normal E and R values for a patient of the specified size. Should the specified IBW value be for a much smaller patient, the first few breaths may be overassisted.
2. Enter endotracheal or tracheostomy tube size. This input is also quite important because part of the delivered assist is based on stated tube size.
3. Set expiratory sensitivity (Esense) to 3 (default value).
4. Set triggering to flow modality at 3 L/min.
5. Enter humidifier volume if applicable.
6. Set high tidal volume limit (VT_I limit) to a value corresponding to 15 mL/kg IBW.
7. Set high peak pressure (HIP) limit to 40 cm H₂O (default). The ventilator caps the assist at the HIP limit less 5 cm H₂O or 35 cm H₂O, whichever is less (i.e., breath is not terminated, but pressure cannot increase beyond this level). Setting HIP to a higher value has no effect in this option. Setting it to a lower value may restrict the assist given unnecessarily.
8. Set percent assist to 70%. It is better to start on the low side and increase if necessary. Most patients do very well at 70%. The use of higher values at the outset, before enough mechanics values have accumulated, may result in overassist.
9. Set PEEP to 5 cm H₂O unless a higher value is deemed necessary for oxygenation.
10. Activate the PAV mode.

Immediate Responses Following Transition to Proportional-Assist Ventilation

The immediate response following a switch to PAV varies considerably depending on what the ventilator settings were before the switch and, in particular, on whether the patient was overassisted or there was significant nonsynchrony.

In patients who are not overassisted (i.e., have good triggering efforts) and have no ineffective efforts before the switch, ventilator rate will remain the same (Fig. 12-23) or increase slightly immediately (i.e., first one to two breaths) following the switch. An immediate increase in respiratory rate in this case (i.e., no ineffective efforts before) is the result improved expiratory asynchrony, with the ventilator cycle terminating soon after the end of effort, as opposed to continuing well into the expiratory phase before the switch (see, e.g., Fig. 12-13). Tidal volume typically decreases immediately, because the ventilator uses normal resistance and compliance values in the first breath. Tidal volume gradually increases as the actual respiratory mechanics are obtained from data during the brief inspiratory plateaus (see Figs. 12-14 and 12-23). Tidal volume often continues to increase for a minute or so before stabilizing. The final tidal volume and respiratory rate are generally not different from PSV unless the patient was overassisted on PSV. In the latter case, V_T and V_E may be slightly lower.

Figure 12-23

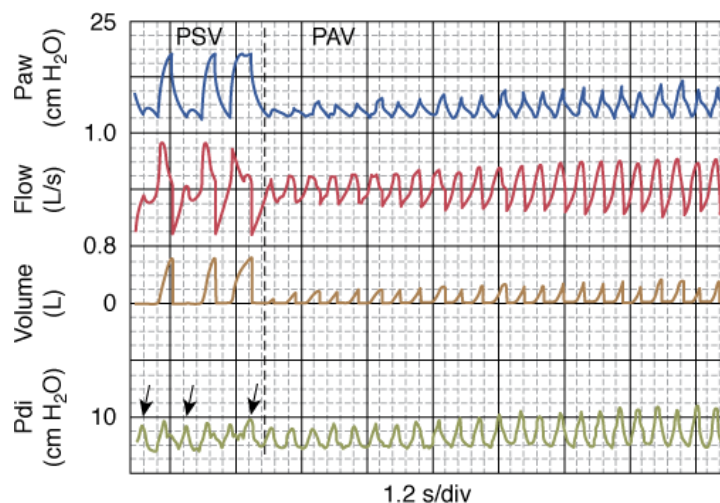


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Example of transition to PAV from PSV in a patient with good efforts (note triggering artifact during PSV) and no important nonsynchrony (no ineffective efforts). Note the brief plateau (arrow) in the first four breaths, used to determine respiratory mechanics. Assist increases gradually as the values of resistance and compliance are updated. Breathing pattern is ultimately similar to the pattern on PSV but the patient is now in a mode that will automatically adjust the assist if there are changes in ventilatory demand or in respiratory mechanics.

In patients with ineffective efforts before the switch there is an immediate increase in ventilator rate as a result of immediate disappearance of the ineffective efforts. This can be dramatic (Fig. 12-24). Concurrently, V_T decreases, often markedly so, in part because resistance and elastance are underestimated initially, but mostly because the inspiratory phase of the ventilator is much shorter than before as a result of disappearance of the preexisting expiratory asynchrony (see Fig. 12-13). V_T then gradually increases, in part because the estimated resistance and compliance become closer to actual values. Because, however, excessive ineffective efforts before the switch generally reflect weak efforts⁷⁷ and overassist, P_{CO_2} and efforts gradually increase. The final breathing pattern is invariably rapid and shallow relative to what it was before. This is so because V_E in assist modes cannot exceed the level dictated by the metabolic hyperbola and the apneic threshold. For example, in a patient whose metabolic hyperbola is similar to that in Figure 12-9 ($VCO_2 = 0.2 \text{ L/min}^{-1}$ and $V_D/V_T = 0.4$) and an apneic threshold of 40 mm Hg, V_E cannot exceed 7 L/min^{-1} or the patient becomes apneic. Assume that before the switch V_T was 0.6 L and ventilator rate was 10 min^{-1} , for a V_E of 6 L/min^{-1} . The underlying respiratory rate, however, was 30 min^{-1} , with 20 ineffective efforts per minute. Upon switching to PAV ventilator rate increases to 30 min^{-1} . The highest that V_T can be without apnea is $7/30$ or 0.23 L.

Figure 12-24

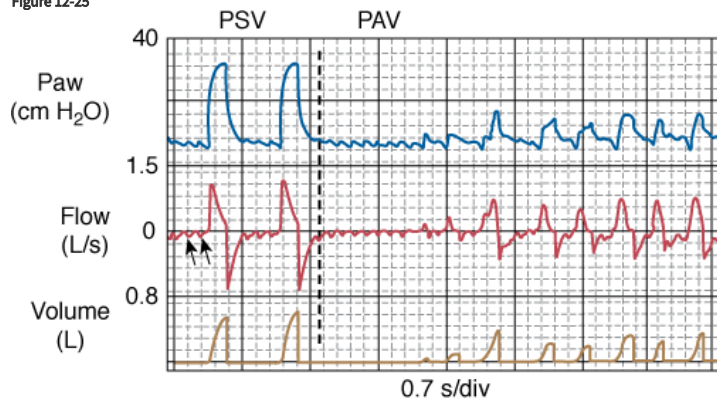


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Example of transition to PAV from PSV in a patient with numerous ineffective efforts on PSV (*arrows*). Note the immediate increase in ventilator rate despite no change in patient's respiratory rate, reflecting the disappearance of ineffective efforts. Tidal volume decreases immediately and rises slowly as the values of resistance and compliance are updated and Pdi increases. Within 45 minutes, tidal volume had increased to approximately half of what it had been on PSV and now ventilator rate is 32 min^{-1} , twice the rate on PSV. Note, however, that despite a modest increase in Pdi, the patient's respiratory rate is actually lower than it was on PSV (36 min^{-1}). The relative rapid shallow breathing in this case reflects better synchrony and not distress. *Paw*, airway pressure; *Pdi*, transdiaphragmatic pressure.

Finally, some patients are so over-assisted on conventional assist modes that they are apneic (e.g., Fig. 12-25). The ventilator continues to be triggered either by cardiac flow artifacts (Fig. 12-25) or by a high backup rate. In such cases, the switch to PAV is immediately followed by apnea (Fig. 12-25). The apnea lasts until rises above the apneic threshold. This may take a minute or more, depending on how much the patient was overventilated before the switch. Flow artifacts do not trigger significant assist in PAV because the inspiratory phase of the artifact is very brief. When breathing resumes, the breathing pattern will obviously be very different from that observed during apneic ventilation.

Figure 12-25

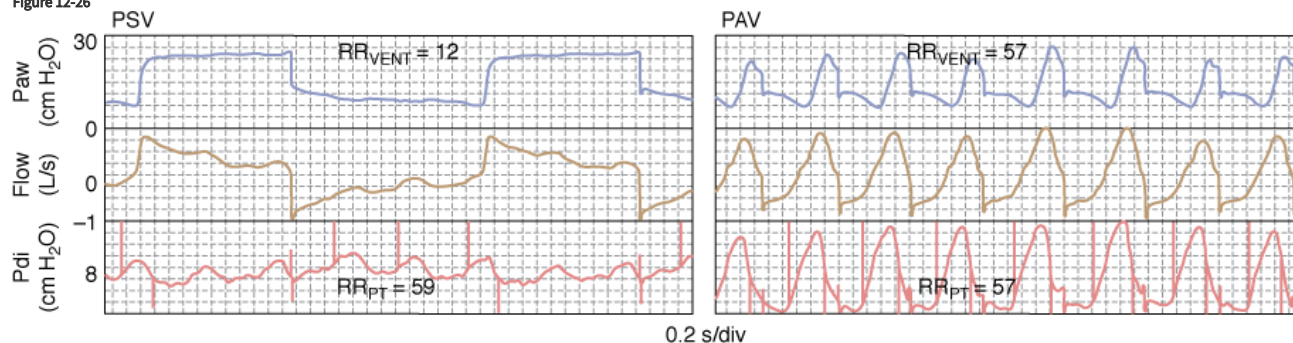


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Example of transition to PAV from PSV in a passive patient in whom large PSV breaths were triggered by cardiac artifacts (*arrows*). Apnea develops upon transition to PAV and continues until PaCO_2 rises above the apneic threshold.

For a physician or therapist who is not familiar with the theory of PAV and the fact that patient-selected breathing pattern can be quite different from what is conventionally viewed as desirable, the above two scenarios (rapid shallow breathing or apnea upon transition to PAV) can be disconcerting, if not alarming. Many would, at this point, give up on PAV and return to the previous conventional mode. This is particularly so when respiratory rate is high (e.g., $>30 \text{ min}^{-1}$) because these rates are conventionally believed to reflect distress. As indicated earlier (see "Respiratory Rate and Breathing Pattern"), the undistressed respiratory rate, defined as the rate observed with high assist that does not increase further at lower levels of assist (see, e.g., Fig. 12-8), can be up to 46 min^{-1} , or even higher (albeit extremely rarely). An extreme example is shown in Figure 12-26. This patient had a respiratory rate of 59 min^{-1} despite high level PSV and weak efforts. His rate did not change upon switching to PAV despite the increase in effort. Clearly, the high rate observed during PSV was not related to distress. If it had been, it would have increased more as respiratory output increased. This patient remained on PAV for several hours and his rate in fact gradually decreased to 50 min^{-1} .

Figure 12-26



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Left panel: Example of extreme tachypnea despite high level of PSV support. Note the efforts are quite small (approximately equal to 4 cm H₂O) and the extreme nonsynchrony (5:1 rhythm). **Right panel:** Upon switching to PAV patient's respiratory rate does not change despite the fact that efforts are now much higher.

A particularly useful approach to distinguish between pathologic apnea and overventilation, and between distress-induced tachypnea and undistressed tachypnea in such cases, is to return the patient to his or her previous settings. If the immediate response upon switching to PAV is apnea, the clinician should decrease the backup rate or increase trigger threshold, as the case may be. Apnea will almost always be observed. Reduce the assist until clear triggering efforts appear. A repeat switch to PAV will now not be followed by apnea, thereby establishing that the apnea was related to overventilation at the previous settings. If the immediate response to PAV is tachypnea, return to the previous settings, slow down the monitor speed to be able to observe eight to ten breaths on the same screen. Then, as the tracing is halfway into the screen, suddenly switch to continuous positive airway pressure. The rate observed in the first two to three breaths on continuous positive airway pressure is the patient's real undistressed rate. It is not possible to develop real distress within a few seconds of removing the assist. If this rate is the same rate observed upon switching to PAV, then one should not be concerned unless, of course, respiratory rate increases further later on. Because patient and ventilator rates are the same with PAV, a further increase in ventilator rate clearly reflects an increase in patient rate. In such cases distress must be suspected and dealt with (see "[Subsequent Management and Troubleshooting](#)" below).

In summary, because of the highly variable immediate responses upon switching to PAV, it is reasonable to wait a few minutes, or perform the above check, before concluding that PAV is not appropriate for the patient. Within a few minutes the body's homeostatic mechanisms will have adjusted respiratory output to the level and pattern preferred by the control system. In patients in whom breathing is rapid and shallow during this waiting period, it is advisable to increase PEEP by 2 to 3 cm H₂O to mitigate atelectasis. In addition, in such patients, particularly in those with a prior history of CO₂ retention, it is advisable to obtain an arterial sample for blood gases several minutes after the switch to ensure that the pH is acceptable. Respiratory acidosis (PaCO₂ > 50 mm Hg and pH < 7.35) with no associated clinical distress suggests respiratory depression. Until the cause of respiratory depression is corrected, the patient should not be treated with PAV.

It is highly advisable to record respiratory rate in the first minute after a switch to PAV, as well as the values of estimated compliance and resistance 3 to 4 minutes later, when an adequate number of measurements have been collected. These values will provide important references for subsequent management.

Subsequent Management and Troubleshooting

1. Subsequent management depends on whether respiratory distress develops within several minutes after the switch and whether there is a trend for respiratory rate or resistance to go up or for compliance to go down.

Given the wide range of documented undistressed respiratory rate in ventilated patients (up to 46 min⁻¹),^{33,34,96,117} it is my opinion that tachypnea with no other supporting manifestations should not be considered as evidence of distress. In their large study comparing PAV with PSV over a 48-hour interval, Xirouchaki et al⁹⁶ defined distress as the presence of at least two of the following: (a) heart rate greater than 120% of the usual rate for longer than 5 minutes and/or systolic arterial blood pressure greater than 180 or less than 90 mm Hg and/or systolic arterial blood pressure changes greater than 20% of the previous value for longer than 5 minutes; (b) respiratory rate greater than 40 breaths/min for longer than 5 minutes; (c) marked use of accessory muscles; (d) diaphoresis; (e) abdominal paradox; and (f) marked complaint of dyspnea. Thus, patients with a respiratory rate greater than 40 breaths/min were maintained on PAV for up to 2 days if there were no other manifestation of distress.

While a high respiratory rate at a given point is not, per se, indicative of distress, an increase in ventilator rate (or in the respiratory frequency to tidal volume [f/V_T] ratio^{127,128}) while the patient is on PAV, provided it is not short-lived, strongly suggests impending failure even in the absence of clinical distress, and even if the absolute rate is modest. Ideally, ventilators should be able to display trends in various physiologic variables (e.g., respiratory rate, compliance, resistance, and so on). In the absence of such a feature one can utilize the respiratory rate obtained immediately after transition to PAV as the reference.

One advantage of PAV is that it makes it possible to monitor resistance and compliance in real time. At least in COPD patients, a trend of increasing resistance and decreasing compliance heralds frank failure.^{1,129} In other patients, the compliance value can be used to derive the Integrative Weaning Index ($\text{compliance} \times \text{arterial oxygen saturation} \times f/V_T$), which shows promise in predicting failure.¹³⁰

2. With the above guidelines, in most patients there will be no clinical distress, and respiratory rate and compliance and resistance will be stable or improving relative to the early measurements (*left blocks* in Fig. 12-22). For these patients, subsequent management is similar to that for other modes and consists of gradual reduction in assist as warranted by clinical condition. So long as there is no clinical distress and respiratory rate and mechanics remain stable or improve, percent assist can be reduced in 10% to 20% decrements every 2 hours until (a) there is no distress or deterioration in respiratory rate and mechanics at 10% to 20% assist. Here, the patient should be considered for extubation as 20% assist represents minimal assist. (b) The patient develops clinical distress or the patient's respiratory rate (or f/V_T) increases or the mechanics deteriorate. In such patients, the assist is increased to the previous level. Such patients are clearly not ready for extubation. Assist should be decreased in smaller steps over longer intervals.
3. A minority of patients will develop distress, or their mechanics will deteriorate, within a few minutes after switching to PAV at 70% assist (*right blocks*, Fig. 12-22). The most common reason for distress at such a high assist (70%) is delayed triggering secondary to severe dynamic hyperinflation or severe respiratory muscle weakness (see “Dynamic Hyperinflation” above). For this reason, the first step is to titrate PEEP up in small increments guided by the results of compliance. Allow 3 minutes between PEEP increases to allow the ventilator to obtain an adequate number of measurements on the new level. If dynamic hyperinflation is in part, or in total, related to expiratory flow limitation, compliance should increase as PEEP is increased. There is no value to increasing PEEP once compliance stabilizes. If distress disappears, the patient is managed with these settings and weaning can then be continued in small steps over long intervals, as tolerated (Fig. 12-22).
4. Distress continues despite PEEP optimization: This scenario is quite uncommon. The caregiver may decide at this point to switch to another mode or to engage in a process of upward titration of percent assist, in 5% steps, with close monitoring of the graphics. In the latter case:

Ensure that the high inspiratory pressure limit is 40 cm H₂O. Inspect the Paw waveform display for several breaths.

- a. If Paw is less than 35 cm H₂O in all breaths, you may continue increasing percent assist. If distress disappears, continue as in the previous groups.
- b. If Paw reaches 35 cm H₂O in the latter part of inspiration only in the occasional breath (see Fig. 12-17), there is volume runaway. Here the problem is almost certainly severe dynamic hyperinflation not responsive to PEEP (i.e., high resistance is in the tubing [including the endotracheal tube]) or severe muscle weakness. If distress is still present, such patients cannot be supported adequately with PAV unless triggering is linked to onset of effort (see “Dynamic Hyperinflation” above and Fig. 12-7). This is the maximum that the patient can be supported in the PAV mode. Switch to another mode.
- c. If Paw reaches 35 cm H₂O in the latter part of inspiration in most breaths, the patient needs more assist than can be provided by the PAV+ option. This scenario indicates high-end nonlinearity (see “Nonlinearity in the Pressure–Volume Relationship within the Tidal Volume Range” above) in a patient with high ventilatory demand. The patient needs to be sedated. If heavy sedation must be used, consider switching to another mode (e.g., assist control).
- d. If Paw reaches 35 cm H₂O *immediately* after triggering and stays there until the cycle ends (i.e., square wave pattern), flow runaway exists. If ATC is concurrently used, it should be discontinued (Fig. 12-19 and related text). If the problem continues and the patient is still in distress, switch to another mode. This problem is occasionally seen in patients with gasping breathing and cannot be corrected by ventilator adjustments.

Important Unknowns

Whereas the physiologic advantages of PAV have been proven, it is not clear whether these necessarily translate into clinical benefits. There are reasons to believe that clinical outcome will improve (see “Clinical Outcome” above). This needs to be confirmed, however. Questions may be of a general nature (e.g., Are mortality, length of intubation, and length of ICU stay less with PAV?) or may be directed at specific aspects that we know affect outcome (e.g., by comparison with optimal protocols with other modes):

If sedation is used on an “as-needed” basis, will patients need less sedation on PAV?

Will tidal volume over the course of illness be smaller on average?

Is weaning faster?

I believe that two other questions need to be addressed. These relate to management of tachypnea that is unrelated to distress, a phenomenon that is evident only when PAV is used:

If a patient decides to breathe in a rapid, shallow manner while on high PAV support (see, e.g., Fig. 12-5B), should the patient be left on PAV or switched to another mode?

If tachypnea (e.g., >35 breaths/min) with no other signs of distress is present during a weaning trial and does not decrease on high PAV, does the patient need to stay on a ventilator?

The Future

With the inclusion of automatic mechanics and “smart” alarms, PAV should become the easiest mode to set and use. There is only one variable to adjust: percent assist. The mode adapts automatically to changes in ventilatory demand and respiratory mechanics. A problem will remain, however, in the management of patients with severe DH secondary to very high expiratory resistance and high ventilatory demand. Future PAV delivery systems should include algorithms to begin the assist at the onset of inspiratory effort.

It is also possible that adjustment of percent assist can be automated. PAV offers two features that would facilitate the complete automation of ventilator adjustment. First, ventilator rate, something ventilators can keep track of easily, is an accurate reflection of the patient’s respiratory rate. Thus, ventilator rate may be used as one feedback element for automatic adjustment of percent assist. Second, because PAV makes it possible to obtain passive mechanics on an ongoing basis, it is also possible to monitor respiratory muscle output and the work of breathing continuously. These can be used as additional feedback signals to automatically adjust percent assist.

Summary and Conclusion

PAV represents a paradigm shift in mechanical ventilation in that control of ventilator output is shifted from the caregiver to the patient. This shift has several advantages in that the ventilator output is more synchronous with patient efforts, and the support adjusts automatically to changes in ventilatory demand and respiratory mechanics. This mode is also the only mode that makes it possible to monitor respiratory mechanics in real time during assisted ventilation. These features have been all been documented in numerous physiologic studies. Until recently, application of PAV in the clinical setting has been hampered by lack of suitable commercial systems and several technical limitations that made it difficult for any but the sophisticated physiologist to use it, and then only for short periods. A commercial ventilator was introduced a few years ago that, in addition to providing the mode, has addressed several of the technical limitations. Now, it is possible to engage in clinical trials that address the important issue of whether better physiology leads to better clinical outcome? PAV is slowly gaining acceptance among clinicians and the main obstacle remains unfamiliarity with the responses to this mode and long-standing dogmas about how breathing should be during mechanical ventilation. It is hoped that this latest description will help in this respect.

Acknowledgments

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